
AUTOMATED INTERPRETABILITY METRICS DO NOT DISTINGUISH TRAINED AND RANDOM TRANSFORMERS

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ABSTRACT

Sparse autoencoders (SAEs) are widely used to extract sparse, interpretable latents from transformer activations, yet it remains unclear whether the metrics used to certify them actually track learned computation. This paper makes three contributions. First, we subject SAE evaluation to a null-model sanity check, asking whether commonly used quality metrics and automatic explanation pipelines can tell trained transformers apart from randomly initialized ones (e.g., where parameters are sampled i.i.d. from a Gaussian). Second, we show empirically that they often cannot: across a wide range of Pythia model sizes and multiple randomization schemes, SAEs trained on randomly initialized transformers produce auto-interpretability scores and reconstruction metrics that are frequently similar to those from trained models. Our central claim follows directly – high aggregate auto-interpretability scores do not, by themselves, guarantee that learned, computationally relevant features have been recovered. Third, we take a first step toward a remedy: a token-distribution entropy measure that begins to expose the feature ‘abstractness’ that aggregate scores miss, and, through toy models, candidate mechanisms for why random networks look interpretable at all. We therefore recommend treating common SAE metrics as useful but insufficient proxies for mechanistic interpretability and adopting routine randomized baselines.

1 INTRODUCTION

Sparse autoencoders (SAEs) have become a popular tool in mechanistic interpretability research, aiming to disentangle the internal representations of neural networks by learning sparse, interpretable features from network activations (Elhage et al., 2022; Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). The recipe is simple: an autoencoder with a high-dimensional hidden layer is trained to reconstruct activations while enforcing sparsity (Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024), in the hope of recovering the underlying concepts or ‘features’ learned by the network (Park et al., 2023; Wattenberg and Viégas, 2024). Crucially, progress in developing better SAEs is measured by quantitative evaluation metrics – above all auto-interpretability scores, which quantify the agreement between generated explanations and activation patterns (Bills et al., 2023; Paulo et al., 2024; Karvonen et al., 2024a). Because these metrics steer the field, whether they actually track learned computation is the question this paper sets out to answer.

For an interpretability method to be considered robust, its evaluation metrics should distinguish features learned through training from artifacts arising from the data or model architecture. A key sanity check is therefore to compare the method’s output on a trained model against a strong null model, such as one with randomly initialized weights (Adebayo et al., 2020). Our first contribution is to bring this sanity check to bear on SAEs: we systematically apply it across a wide range of Pythia model sizes and several distinct randomization schemes, treating the randomized transformer as an explicit null model for SAE evaluation. Our second, and central, contribution is the empirical finding that several common quantitative metrics fail this check – they do not always clearly distinguish between the trained and randomized settings. In particular, we find that SAEs trained on transformers with random parameters can yield latents whose auto-interpretability scores (Bills et al., 2023; Paulo et al., 2024) are surprisingly similar to those from a fully trained model.

The headline consequence of this finding is a caution about what these metrics of SAE quality can tell us: high auto-interpretability scores alone do not guarantee that an SAE has identified complex,

learned computations. Instead, such scores may sometimes reflect simpler statistical properties of the training data (Dooms and Wilhelm, 2024) or architectural inductive biases that are present even without training. Indeed, one could argue that a randomly initialized network still performs a basic form of computation, such as preserving or amplifying the sparse structure of its inputs (Section 4). From this perspective, SAEs might faithfully interpret this simple, inherent computation – which is precisely why a metric that scores it highly cannot, on its own, be read as evidence of learned structure.

We stress that this is a statement about the metrics, not about SAEs: while some SAE features from trained models clearly arise from learned computation, the commonly used aggregate metrics are often insufficient for determining whether a given SAE has learned these more complex features. The practical implication for mechanistic interpretability research is twofold, and constitutes our third contribution: more rigorous methods are needed to distinguish artifacts from genuinely learned computations, and interpretability techniques should be routinely validated against appropriate null models. As a concrete first step toward such a method, we introduce a token-distribution entropy measure that, unlike aggregate auto-interpretability scores, begins to separate the simple single-token features of randomized models from the more abstract features of trained ones (Section 3).

Finally, we take up the question of *why* random transformers look interpretable at all, offering candidate mechanisms rather than a definitive answer. At a high level, there are two hypotheses: (1) the input data already exhibits superposition, and randomly initialized neural networks largely preserve this superposition; and (2) randomly initialized neural networks amplify or even introduce superposed structure to the input data (e.g., given dense input generated i.i.d. from a Gaussian). We present toy models to demonstrate the plausibility of both hypotheses in Section 4, but defer conclusions as to which mechanism predominates to future work.

Contributions In summary, this paper contributes the following.

- We frame randomly initialized transformers as an explicit null model for SAE evaluation and run this sanity check systematically across Pythia models from 70M to 7B parameters and multiple randomization schemes (Section 3).
- We show empirically that widely used SAE quality metrics – most notably aggregate auto-interpretability scores, as well as reconstruction-based metrics – are frequently similar for trained and randomized transformers, and are therefore insufficient on their own to certify that learned computation has been recovered (Section 3).
- We propose token-distribution entropy as a proof-of-concept measure of feature ‘abstractness’ that surfaces a trained-vs-randomized distinction the aggregate metrics miss, and we recommend routine randomized null-model baselines for interpretability research (Sections 3 and 6).
- We provide toy models offering two candidate mechanisms – preservation and amplification of input superposition by random networks – for why these patterns arise (Section 4).

2 RELATED WORK

Sparse dictionary learning Under a different name, ‘superposition’ in visual data is one of the foundational observations of computational neuroscience. Olshausen and Field (1996; 1997) showed that the receptive fields of simple cells in the mammalian visual cortex can be explained by sparse coding: representing a relatively large number of signals (sensory information) by simultaneously activating only a small number of elements (neurons). Viewed through this lens, efforts to extract the ‘underlying signals’ responsible for neural network activations are naturally cast as a coding-theoretic problem (Marshall and Kirchner, 2024).

Sparse dictionary learning (SDL) formalizes this problem by approximating a set of input vectors as linear combinations of a relatively small, usually overcomplete, set of learned basis vectors – one of higher dimension than the inputs. Classical SDL algorithms include Independent Component Analysis (ICA), which finds a linear representation of the data whose components are maximally statistically independent (Bell and Sejnowski, 1995; Hyvärinen and Oja, 2000). Sparse autoencoders (SAEs) are a simple neural-network instance of SDL (Lee et al., 2006; Ng, 2011; Makhzani and Frey, 2014): an autoencoder with a single hidden layer many times larger than the input activation vectors

is trained with an objective that imposes or incentivizes sparsity in its hidden-layer activations, in order to recover this structure. Throughout, we call a single neuron (dimension) in the autoencoder’s hidden layer a **latent**.

Mechanistic interpretability Recently, it has become common to understand ‘features’ or concepts in language models as low-dimensional subspaces of internal model activations (Park et al., 2023; Wattenberg and Viégas, 2024; Engels et al., 2024). Where such sparse or ‘superposed’ structure exists, one expects to be able to ‘intervene on’ or ‘steer’ the activations – modifying or replacing them to express different concepts and thereby influence model behavior (Meng et al., 2022; Zhang and Nanda, 2023; Heimersheim and Nanda, 2024; Makelov, 2024; O’Brien et al., 2024).

SAEs have become the dominant approach for discovering such features, typically by training a single autoencoder to reconstruct the activations of a single network layer, e.g., the transformer residual stream (Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). The many SAE variants proposed to date differ mainly in the activation function applied after the linear encoder (Makhzani and Frey, 2014; Gao et al., 2024; Rajamanoharan et al., 2024b; Lieberum et al., 2024), but also in their training objectives (Braun et al., 2024; Farnik et al., 2025) and in whether they are applied to one layer or several simultaneously (Yun et al., 2021; Lawson et al., 2024; Lindsey et al., 2024).

How to *evaluate* these SAEs is the concern most directly relevant to our work. Beyond reconstruction error and preservation of the underlying model’s performance, SAEs have been judged by whether they capture specific concepts (Gurnee et al., 2023; Gao et al., 2024) or factual knowledge (Huang et al., 2024; Chaudhary and Geiger, 2024), and whether the resulting features support ‘unlearning’ of concepts (Karvonen et al., 2024b).

Automatic neuron description Because SAEs often learn tens of thousands of latents, describing them by hand is infeasible. Early work by Yun et al. (2021) found the tokens that maximally activate a dictionary element over a text dataset and inspected the activation patterns manually. The now-standard alternative is to collect latent activation patterns over a text dataset and prompt a large language model to explain them (e.g. Bills et al., 2023; Foote et al., 2023) – a recipe that has been widely adopted (e.g. Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024) and whose output metrics are exactly what we probe.

The auto-interpretability (as in self-interpreting) paradigm originates with Bills et al. (2023), who generate an explanation for a language-model neuron’s activation patterns over dataset examples, simulate the patterns from the explanation, and score the explanation by comparing observed and simulated activations. Paulo et al. (2024) make this practical, introducing classification-based fidelity measures that are inexpensive to compute relative to full simulation, together with an open-source pipeline. Choi et al. (2024) further scale explanation quality via best-of- k sampling over different subsets of maximally activating examples; notably, they fine-tune Llama-3.1-8B-Instruct on the top-scoring explanations to obtain inexpensive ‘explainer’ and ‘simulator’ models.

Polysemanticity Lecomte et al. (2024) noted that neurons may become polysemantic incidentally. A polysemantic neuron (basis dimension) of a network layer represents multiple interpretable concepts (Elhage et al., 2022; Scherlis et al., 2023); unsurprisingly, individual neurons in a randomly initialized network may already be polysemantic. It is worth being precise about scope here, because it matters for our claims: our work studies *superposition* (Elhage et al., 2022; Chan, 2024), a property of the representations learned across a whole network layer rather than of any individual neuron. Superposition lets a layer as a whole represent more (sparse) features than it has (dense) neurons through sparse coding – only a few concepts are active at a time, i.e., at a given token position.

Training only the embeddings Zhong and Andreas (2024) showed that transformers acquire surprising algorithmic capabilities when only the embeddings are trained and no other parameters, demonstrating that the behavior of a randomly initialized transformer can be shaped considerably by training just a few parameters. Our setting is deliberately different, and the distinction is important for interpreting our null model: beyond considering SAEs, we randomize *all* parameters, including the embeddings, in our ‘Step-0’ and ‘Re-randomized incl. embeddings’ variants. The ‘Re-randomized excl. embeddings’ variant does retain pre-trained embeddings, but – unlike that setting – we never train around them: we freeze the pre-trained embeddings and randomize every other weight (Section 3).

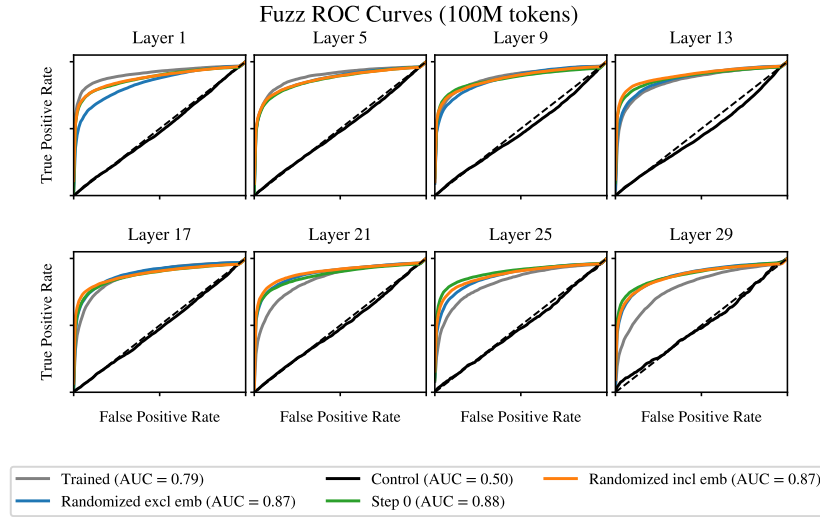


Figure 1: Our central finding at a glance: ‘fuzzing’ ROC curves vs. layer for Pythia-6.9b (100 latents sampled per SAE). The trained model (gray line) and the randomized variants (colored) overlap almost entirely, while only the control (black) drops to near chance (dotted). The takeaway is that aggregate AUROC alone cannot attribute latents to learned computation – it fails to separate trained from randomized transformers. See Figure 2 for other metrics and model sizes, and Appendix E for multiple random seeds.

Random transformers for board games Karvonen et al. (2024c) found that SAEs extracted meaningful structure from chess games far more effectively with pre-trained transformers than with random-weight ones – seemingly the opposite of our finding. The apparent tension resolves once the data is considered: board-game data is very unlike language data. Language has good reason to be sparse (a concept such as ‘serendipitous’ appears only rarely) and to have that sparse structure ‘aligned’ with conceptual meaning; board games need not share this property, since a useful concept such as a knight fork does not necessarily occur sparsely. This contrast helps delimit the scope of our claims: they concern language data specifically.

Random one-layer transformers Bricken et al. (2023) found that auto-interpretability scores discriminated effectively between random and trained one-layer transformers. Our results refine rather than contradict this: auto-interpretability scores for randomized models were relatively low for smaller models (e.g., Pythia-70m) but the gap narrowed for larger models (e.g., Pythia-6.9b), which is precisely where the metrics stop being reliable.

3 RESULTS

This section presents our first two contributions: the null-model experimental design, and the empirical finding that standard SAE metrics fail to separate trained from randomized transformers. The setup is deliberately broad so that the finding cannot be dismissed as an artifact of one model or one form of randomization. We trained per-layer SAEs on the residual-stream activation vectors of transformer language models from the Pythia suite, spanning 70M to 7B parameters (Biderman et al., 2023). Our comparison holds the SAE training procedure fixed while varying the underlying transformer across five carefully constructed variants:

- **Trained:** the usual, trained model – our reference point.
- **Re-randomized incl. embeddings:** every model parameter, embeddings included, is re-initialized by sampling Gaussian noise whose mean and variance match those of the corresponding original, trained weight matrix.

- **Re-randomized excl. embeddings:** as above, but the embedding and unembedding matrices are left untouched, i.e., identical to the trained model.
- **Step-0:** the Pythia step0 revisions, i.e., the original model weights at initialization, before any learning (Biderman et al., 2023).
- **Control:** the original, trained model, but with input token embeddings replaced at inference time by i.i.d. standard Gaussian noise drawn afresh for each token, so that a token has no consistent embedding vector. This is our lower bound: we expect auto-interpretability here to perform at chance.

Together, the four non-trained variants act as null models spanning full re-randomization, partial re-randomization, the true initialization state, and a chance-level control, giving us a stringent test of whether the metrics respond to learning.

For our primary experiments, we trained SAEs on 100M tokens from the RedPajama dataset (Weber et al., 2024) with a 10M-token activation buffer; Appendix C confirms that a subset of these results is unchanged when SAEs are instead trained on one billion tokens, ruling out insufficient data as an explanation. To keep compute tractable at scale, we trained an SAE at every layer for models below 410M parameters, at every second layer for Pythia-1b, and at every fourth layer for Pythia-6.9b.

Unless otherwise stated, we trained k -sparse autoencoders (TopK SAEs; Makhzani and Frey 2014; Gao et al. 2024) with expansion factor $R = 64$ and sparsity $k = 32$. So that our headline finding cannot be attributed to a particular hyperparameter choice, we verified its robustness on Pythia-160m across expansion factors equal to powers of 2 from 16 to 128 and sparsities of 16 and 32 (Figure 18). Our training implementation builds on Belrose et al. (2025), and our evaluations on Caden et al. (2025) and Karvonen et al. (2024a).

Auto-interpretability We first fix the primary metric under test. A feature explanation that names a concept can be handed to a classifier that predicts whether the concept appears in a text input, and that classifier can be scored by standard metrics such as the area under the receiver operating characteristic (ROC) curve (AUROC). Paulo et al. (2024) proposed two such classification tasks – ‘fuzzing’ and ‘detection’. In ‘fuzzing’ scoring, positive and negative token examples for a latent (with non-zero and zero activation, respectively) are delimited with special characters, and a language model is prompted to identify, from the latent’s explanation, which examples have been correctly delimited. In ‘detection’, a language model instead identifies which examples contain activating tokens for each feature. The original ‘simulation’ scoring of Bills et al. (2023), based on the correlation between predicted and observed activations, is more expensive to compute.

Except where noted, we report ‘fuzzing’ scores as our measure of auto-interpretability, since they have been shown to correlate with simulation scoring (Paulo et al., 2024); the corresponding ‘detection’ AUROC curves appear in Appendix B. For each trained SAE (underlying model, variant, and layer), we randomly sampled 100 features to obtain auto-interpretability scores, following the implementation of Paulo et al. (2024). We used the open-source Meta-Llama-3.1-70B-Instruct-AWQ-INT4 model to both generate explanations and make predictions – larger than the 8B models of Choi et al. (2024), and open, unlike Bills et al. (2023).

The result is direct evidence for our central claim: auto-interpretability scores were far more similar between the trained and randomized models than either was to the control (Figures 1 and 2). Because the trained and randomized ROC scores are so close, ‘fuzzing’ auto-interpretability applied to SAE latent explanations does not, on its own, meaningfully distinguish these underlying models – exactly the failure mode our sanity check was designed to detect.

Evaluation To show that the failure is not specific to auto-interpretability, we broadened the analysis to a suite of standard SAE evaluation metrics alongside the auto-interpretability AUROC, for Pythia models from 70M to 6.9B parameters. The same pattern recurs across metrics: the randomly initialized and re-randomized models (Figure 2; blue, green, orange lines) sit much closer to the trained model (gray lines) than to our control (black lines).

Beyond this headline pattern, the plots reveal several secondary regularities worth noting. The control is the clear outlier: its cosine similarity between original and reconstructed activations and its explained variance are often far lower than the other models’, and its reconstruction error tends to rise

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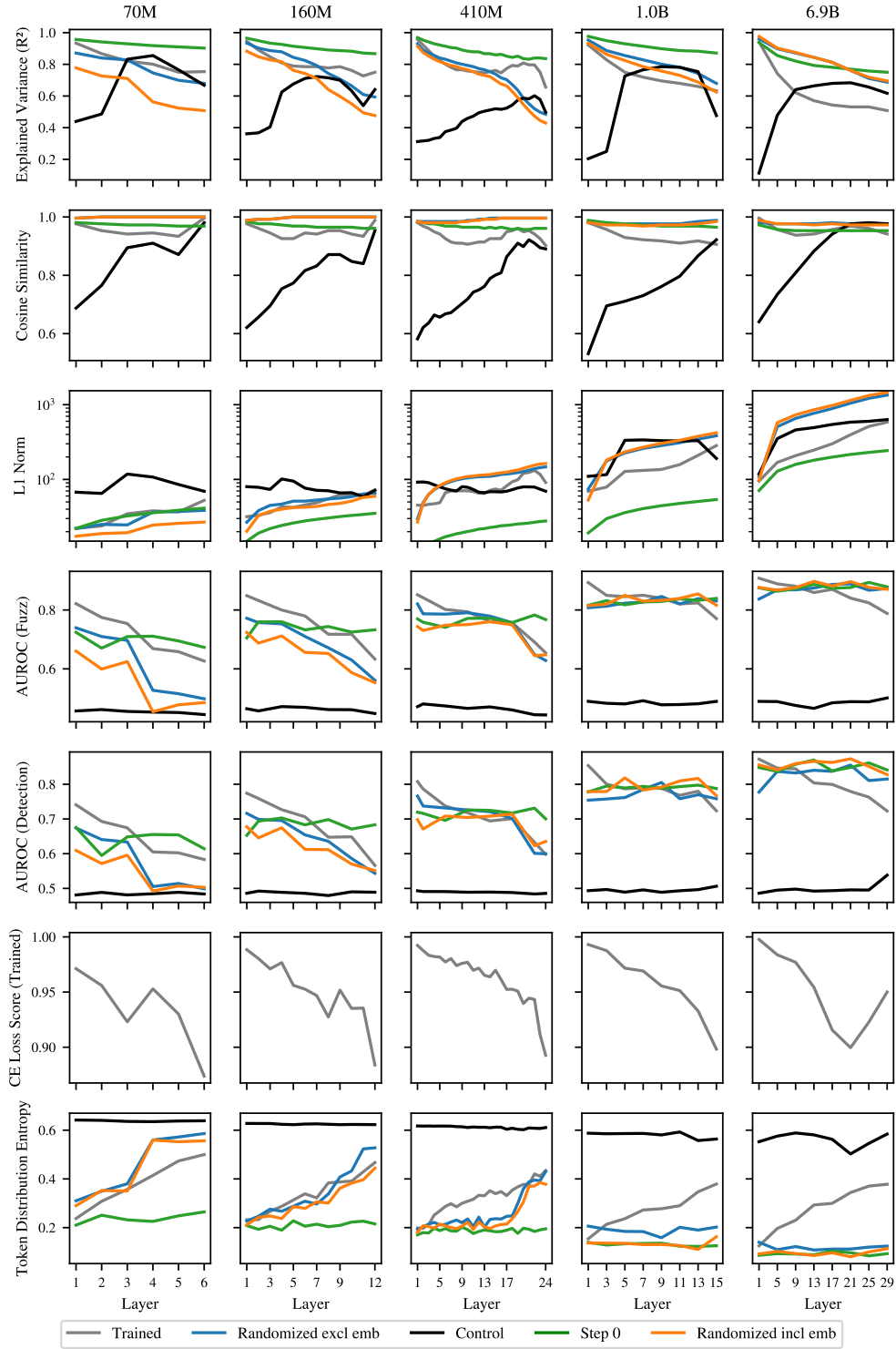


Figure 2: The finding generalizes beyond a single metric and model. Sparse autoencoder performance across Pythia models (70M to 6.9B parameters) for a suite of standard metrics. The key pattern: every variant except the control tracks the trained model closely, and this convergence tightens as model scale grows – despite the variants differing fundamentally in initialization. In short, standard metrics do not separate trained from randomized transformers, whereas the control is clearly distinguishable.

across layers while the other variants’ fall. A plausible reason is that a Gaussian is the highest-entropy distribution with fixed mean and variance (Jaynes, 2003); we speculate that Gaussian vectors are the ‘least structured’ and hence hardest for SAEs to reconstruct. As these Gaussian-distributed activations propagate through successive layers, we would expect them to become less Gaussian and perhaps more ‘sparse’, i.e., easier to reconstruct (Section 4).

A second regularity is that the re-randomized variants (blue and orange lines) resemble the trained model more closely than the variant at initialization (green line), most visibly in the L^1 norm values for larger models. We attribute this to parameter norms, which can differ greatly between a trained model and its state at initialization, whereas our randomization procedure was designed specifically to preserve parameter norms relative to the trained model. Parameter scale across layers plausibly matters, e.g., for controlling the growth of activations along the residual stream (Liu et al., 2020). A third regularity: in the AUROC plots, every variant except the control improves with model size. We speculate that features become more specific as SAE size grows – in smaller SAEs each latent must explain more of the input, making the classification task easier for larger SAEs.

Row five of Figure 2 reports the cross-entropy (CE) loss score, or loss recovered, against model layer: the increase in loss when original activations are replaced by their SAE reconstructions, divided by the increase when they are ablated to zero. Here the ‘trained’ variant SAEs perform on par with others in the literature (e.g., Kissane et al., 2024; Rajamanoharan et al., 2024a; Mudide et al., 2024). This metric is the one clear exception to our pattern, but only trivially so: the CE loss score is meaningful only for the trained variant, because for any randomized variant the loss is very poor regardless of whether original or reconstructed activations are used.

Latent explanation complexity Having shown what the aggregate metrics miss, we now turn to our third contribution: a simple measure that begins to recover the distinction they lose. The motivation is an expectation that the aggregate scores failed to confirm – that SAEs applied to trained vs. randomized transformers should discover qualitatively different features. Specifically, we expected the randomized variants to learn relatively simple features keyed to surface characteristics of the input text, but not the more complex, abstract features characteristic of trained transformers (Templeton et al., 2024). Qualitative evidence for this expectation appears in Appendix J, which shows a random sample of features and their maximally activating dataset examples for each Pythia-6.9b variant, with more detail in Appendix L.

To make the intuition quantitative, we start from an anecdotal observation: a substantial fraction of SAE latents fire on only a single token, or a small number of distinct tokens, across a text dataset (e.g., Lin and Bloom, 2024; Dooms and Wilhelm, 2024). A latent’s ‘token-specificity’ is therefore a natural proxy for its simplicity, and we capture it with the entropy of the observed distribution of latent activations over token IDs – higher entropy means the activations are more ‘spread out’ and the latent is less token-specific. We take this distribution to be the total latent activation per token across the maximally activating examples used to generate auto-interpretability explanations. The per-latent relationship between this entropy and the ‘fuzzing’ AUROC score is given in Appendix H.

The payoff is visible in the last row of Figure 2, which reports this token-distribution entropy and – unlike the aggregate metrics above – pulls the variants apart. The control sits at consistently high entropy, as expected, since a token’s embedding is resampled i.i.d. from a Gaussian on each occurrence and so carries no consistent vector (Section 3). For the trained variant, entropy rises across layers: deeper in the model, a latent’s maximally activating examples are less likely to concentrate on a single token – again as expected, since later layers should host more abstract features less tied to token embeddings. The randomized models are the telling case: their entropy tends to be lower than either the trained or control variant, indicating latents that fire specifically at one or a few token IDs.

The upshot, combining this with the preceding results, is that standard SAE quality and auto-interpretability metrics are blind to an important property of SAE features – their ‘abstractness’. Token-distribution entropy is not a direct measure of abstractness, and we present it as a proof of concept rather than a finished metric; nevertheless it makes the missing distinction legible. Read against their trained-like auto-interpretability scores, the randomized variants are revealed to learn only simple, single-token features, and, crucially, those features do not grow more complex with layer depth as the trained variant’s do.

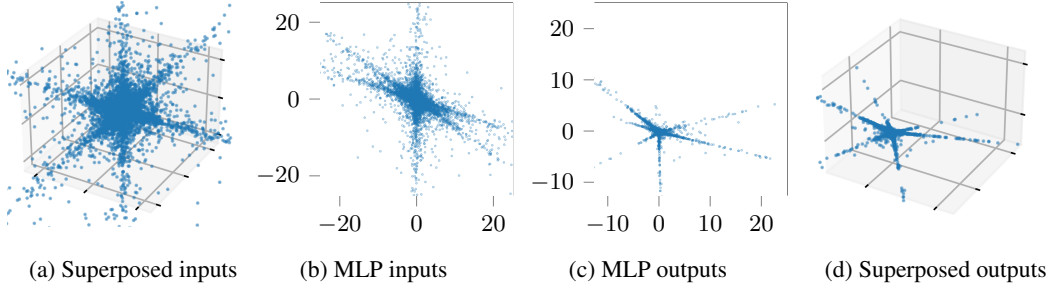


Figure 3: Illustrating mechanism one: a randomly initialized network preserves (and appears to sharpen) superposition in its input. We draw 10K samples of $n_s = 3$ sparse input features from a Lomax distribution with shape $\alpha = 1$ and scale $\lambda = 1$, project them to $n_d = 2$ dense input features by an i.i.d. standard normal matrix, pass the result through a two-layer ReLU MLP with hidden size $4n_d$, and recover $n_s = 3$ sparse outputs via the inverse of that projection matrix. The superposed structure survives the nonlinearity, foreshadowing the amplification we test next.

4 A TOY MODEL OF SUPERPOSITION IN RANDOM NETWORKS

Our final contribution addresses the natural follow-up question: *why* do randomized transformers yield such sparse, interpretable-looking activations? In Section 1 we raised two candidate mechanisms – that the input data already exhibits superposition which neural networks preserve, or that neural networks amplify or even introduce superposition. This section examines both through toy models. We find evidence consistent with each mechanism; we present them as plausible explanations rather than a verdict, and leave the question of which predominates for real randomized transformers to future work.

4.1 MATRIX MULTIPLICATIONS PRESERVE SUPERPOSITION

We begin with the simplest case, which already makes the first mechanism precise: multiplication by a weight matrix W preserves superposition. Suppose superposed input data x is generated by first drawing n_s i.i.d. ‘sparse’ features z from a heavy-tailed Lomax distribution $z \sim \text{Lomax}(\alpha, \lambda)$. We project the higher-dimensional, sparse z down to lower-dimensional, dense x with a matrix D and add small-variance Gaussian noise Σ , so that $x \sim \mathcal{N}(x; Dz, \Sigma)$. The key observation is that multiplying x by any matrix W leaves $x' = Wx$ *also* superposed: it follows the same generative model as x , only with a different noise covariance and a different map from z to x , namely $x' \sim \mathcal{N}(z; WDz, W\Sigma W^T)$.

This intuition plausibly survives the nonlinearities of a real network, which we check by visualizing the effect of passing the dense activations through a simple feed-forward network (MLP). In the example of Figure 3, $n_s = 3$ sparse features are projected to $n_d = 2$ dense features, and the MLP outputs remain visibly superposed despite the nonlinearity. More suggestively, the figure hints at the second mechanism – amplification rather than mere preservation: comparing inputs (Figures 3a and 3b) with outputs (Figures 3d and 3c), fewer points fall between the ‘arms’ of the outputs. We test this amplification hypothesis directly next.

4.2 DO RANDOM NNs PRESERVE OR AMPLIFY SUPERPOSITION?

To test the amplification hypothesis directly, we generated toy data following Sharkey et al. (2022): sampling ground-truth features on a hypersphere and generating correlated feature coefficients such that only a small number are active at once (Appendix I.1). We passed these inputs through a two-layer MLP at initialization and trained SAEs separately on the inputs and on the outputs. The control is Gaussian-distributed input matched in mean and standard deviation to the superposed toy data, and we used standard SAEs with an L^1 sparsity penalty (Appendix I.2).

As a first step, and following Sharkey et al. (2022), we confirmed that SAEs recover the ground-truth features generating the data (Figure 4), measured by the mean max cosine similarity (MMCS): for each data-generating feature we take its maximum cosine similarity with the SAE’s learned features

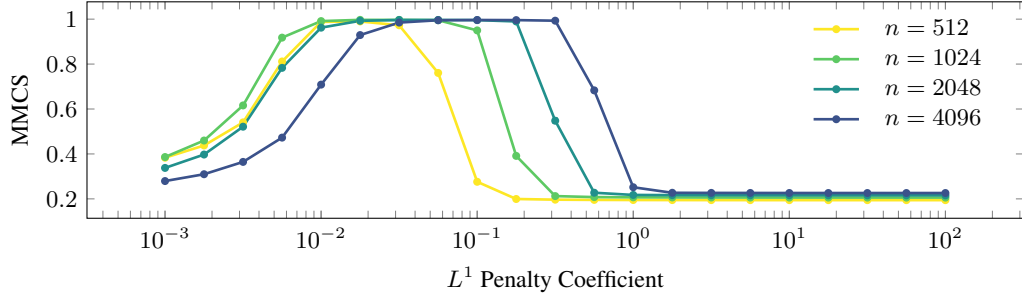
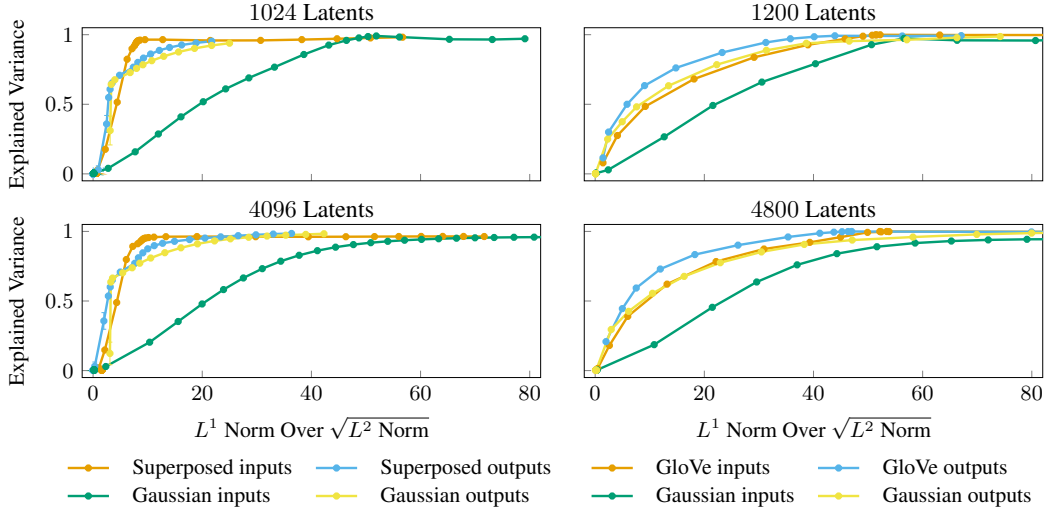


Figure 4: A sanity check on the toy setup: mean max cosine similarity (MMCS) between the features learned by a standard SAE (decoder weight vectors) and the data-generating features, as a function of the L^1 penalty coefficient, following Sharkey et al. (2022). A ‘Goldilocks zone’ exists in which SAEs near-perfectly recover the data-generating features, provided they have enough latents to represent them.



(a) Toy datasets following Sharkey et al. (2022). (b) 300-dim. GloVe vectors (Pennington et al., 2014).

Figure 5: Evidence that random networks sparsify their inputs. Pareto frontiers of explained variance against the L^1 norm divided by the square root of the L^2 norm (sparsity), for datasets that may exhibit superposition, Gaussian controls with matched mean and variance, and the corresponding outputs after passing each through a randomly initialized two-layer MLP (Section 4.2). Each point is one choice of L^1 penalty coefficient; the superposed-vs-control gap shrinks after the MLP, indicating that the network raises the apparent sparsity of even the control inputs.

(decoder weight vectors) and average over data-generating features. The MMCS, however, is only defined for the MLP inputs, where the data-generating features are known; analyzing the MLP outputs requires a different handle. For that, we use an SAE’s ability to reach low reconstruction error at high sparsity as a proxy for how strongly the training data exhibits superposition, sweeping the L^1 penalty coefficient to trace Pareto frontiers of explained variance against sparsity (Figure 5a).

The results support amplification, with a nuance. As expected, SAEs achieve much greater sparsity at a given explained variance for the superposed inputs than for the Gaussian control (Figure 5a; orange and blue-green). The telling comparison is on the outputs: the gap between the superposed outputs (the MLP’s outputs given superposed inputs) and the Gaussian outputs is much smaller, with only slightly greater sparsity at a given explained variance. In other words, the outputs of a randomly initialized MLP are already fairly sparse regardless of the input distribution – consistent with the idea that random networks impose sparse structure of their own. Other sparsity measures and hyperparameters, which tell the same story, are collected in Appendix I.

4.3 DO TOKEN EMBEDDINGS EXHIBIT SUPERPOSITION?

If random networks preserve or amplify superposition, then the last link in the chain is whether the inputs to transformer language models – their token embeddings – exhibit superposition to begin with, which would let both mechanisms explain our main results (Section 3). We test this by applying the Section 4.2 procedure to language data, training SAEs on pre-trained GloVe word vectors, the embedding matrices of Pythia models, the outputs of passing these through a randomly initialized two-layer MLP, and Gaussian controls. The setup is otherwise identical to Section 4.2, except that the number of data points is fixed by the number of word embeddings or tokens and we use a single random seed.

Two observations follow. First, the gap between the Pareto frontiers of the GloVe word vectors and their Gaussian controls (Figure 5b) is smaller than for the toy superposed datasets of Section 4.2 (Figure 5a) – so embeddings are only mildly superposed relative to the idealized toy data. Second, and reinforcing the amplification mechanism, both inputs’ Pareto frontiers improve after passing through a randomly initialized two-layer MLP, again pointing to random networks ‘sparsifying’ their inputs (increasing the apparent degree of superposition).

5 LIMITATIONS

To keep our central claim in proportion, we note its boundaries. Our contribution is to show that auto-interpretability measures can produce apparently meaningful, interpretable results for SAEs trained on randomly initialized models – models unlikely to contain computationally interesting features – and this scope carries two caveats. First, on coverage: since no study can span all datasets and architectures, we deliberately concentrated on the Pythia family, widely adopted in mechanistic interpretability research (e.g., Paulo and Belrose, 2025; Ghilardi et al., 2025; Mueller, 2024), and on RedPajama-V2 as representative pre-training data for language models and SAEs. Second, on the explainer: we used the default explanation model of the EleutherAI auto-interpretability framework (Caden et al., 2025), and alternative models could yield further insight into aggregate behavior and explanation quality. Most importantly, we are careful about what the finding does *not* say: we do not claim that SAEs fail to capture information present in trained transformers beyond what is present in randomized ones. Our claim is narrower and about the metrics – that aggregate auto-interpretability measures do not, by themselves, signal the existence of interesting underlying features.

6 CONCLUSION

We set out to test whether the metrics that steer SAE research actually track learned computation, and our contributions answer that question. Applying sparse autoencoders to both trained and randomly initialized transformers and evaluating them with a suite of common quantitative metrics, our central empirical finding is that, under certain conditions, these metrics – particularly aggregate auto-interpretability scores – are surprisingly similar in both settings. Features derived from trained transformers are, on inspection, qualitatively more complex and abstract, especially in later layers, yet the aggregate metrics routinely fail to register this distinction.

The right reading of this result is a limitation of current evaluation methods, not of SAEs: it does not imply that SAEs on real models fail to learn meaningful computational features. The concrete lesson is that high aggregate auto-interpretability scores are insufficient evidence for the discovery of complex, learned computations, since they may instead reflect simpler structure – inherent in the data or the architecture – that survives even random weights. As a first step past this limitation, our token-distribution entropy analysis, though preliminary, serves as a proof of concept: it surfaced differences in feature ‘abstractness’ that aggregate auto-interpretability scores missed. We therefore leave two recommendations for the field. First, future work should develop more robust metrics that quantify the computational significance of the features SAEs discover, building on measures such as ours. Second, and most actionably, interpretability techniques should be routinely benchmarked against strong, appropriately constructed null models – such as the randomly initialized transformers used here – because without such baselines it is difficult to confidently attribute discovered features to learning.

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A BROADER IMPACT

This work scrutinizes a method currently used for mechanistic interpretability of LLMs, and its results challenge certain assumptions about sparse autoencoders. Our core demonstration – that SAEs can produce similar aggregate auto-interpretability scores for both random and trained transformers – raises important questions about what these SAE evaluation methods actually capture.

By clarifying what the metrics of SAE quality do and do not measure, we hope this work contributes to a better-informed search for improved SAE-like methods, and thereby helps make these models more interpretable and mitigate the harms they might cause. Because our study is empirical and simply shows that a presently used method yields interpretations for both random and trained transformers, we judge the risk of negative societal impact to be minimal.

B AUTO-INTERPRETABILITY ROC CURVES

These curves substantiate, at additional model sizes, the central finding that auto-interpretability fails to separate trained from randomized SAEs. Figures 6, 8, and 12 show the similarity in ‘fuzzing’ AUROC between trained and randomized SAEs for the 70M, 160M, and 1B models. Figures 7, 9, and 13 show the same similarity for ‘detection’ AUROC on the 70M, 160M, and 1B models.

B.1 PYTHIA 70M

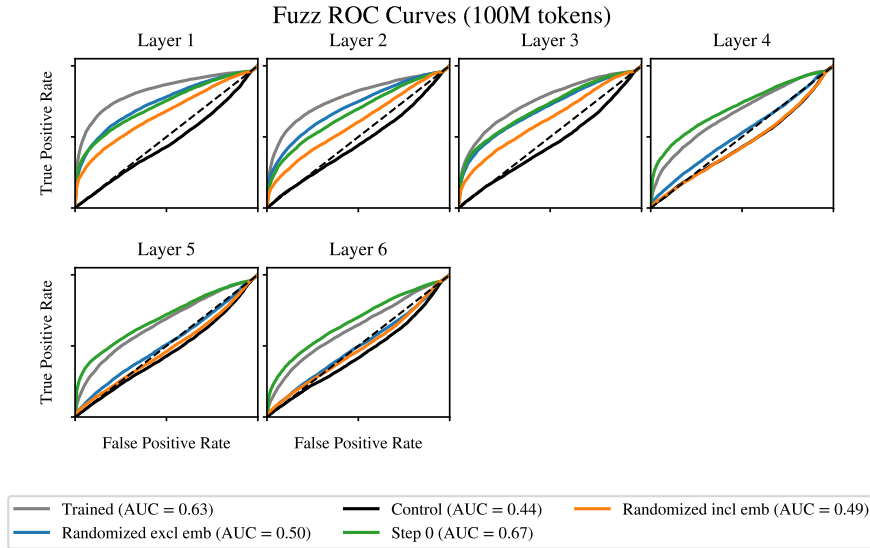


Figure 6: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-70m over 100 SAE latents. Consistent with our main finding, the SAE variants perform similarly; performance also degrades overall as the layer index increases.

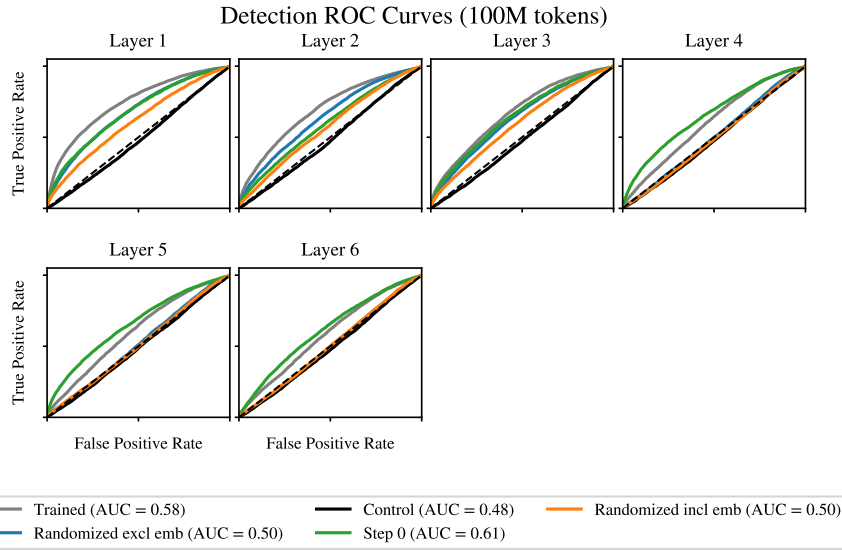


Figure 7: ‘Detection’ auto-interpretability ROC curves for Pythia-70m over 100 SAE latents. As with ‘fuzzing’, the variants perform similarly and performance degrades overall as the layer index increases.

B.2 PYTHIA 160M

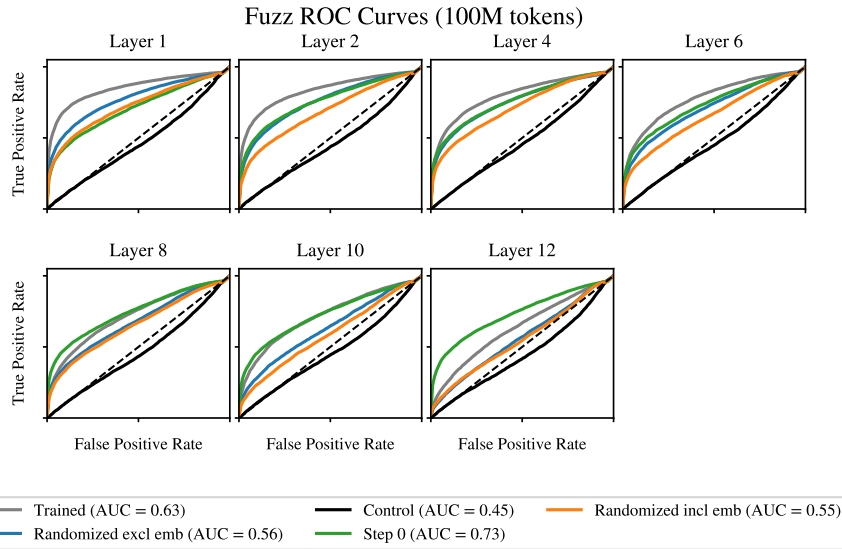


Figure 8: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-160m over 100 SAE latents. Again the SAE variants perform similarly, with an overall degradation as the layer index increases.

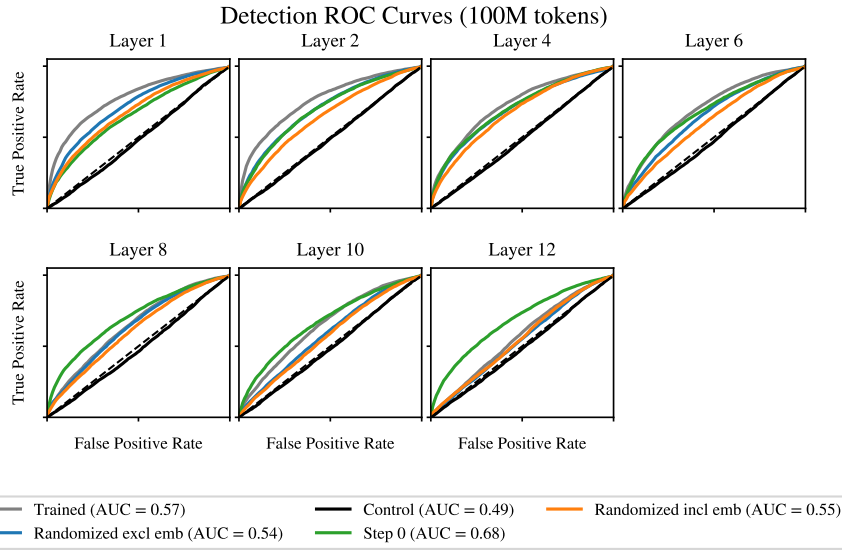


Figure 9: ‘Detection’ auto-interpretability ROC curves for Pythia-160m over 100 SAE latents. The variants again perform similarly, with an overall degradation as the layer index increases.

B.3 PYTHIA 410M

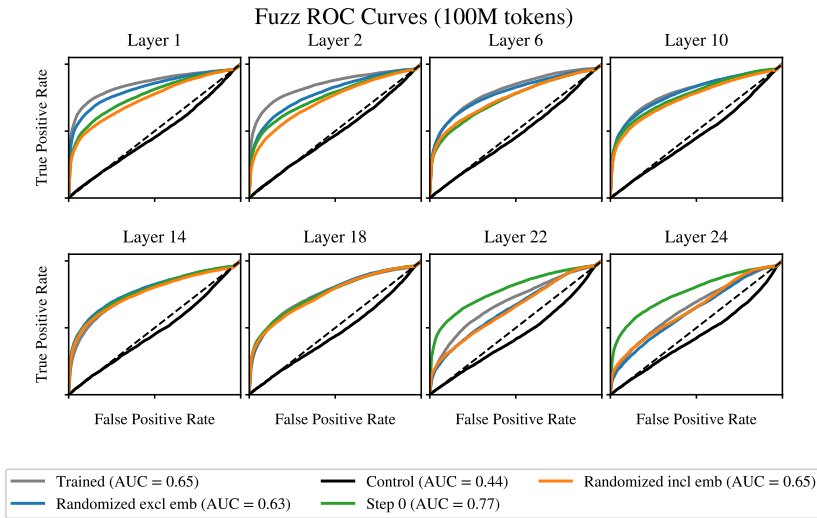


Figure 10: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-410m over 100 SAE latents. The variants perform similarly and degrade overall with layer index; here, too, the auto-interpretability scores fail to distinguish trained from randomized models – our central finding.

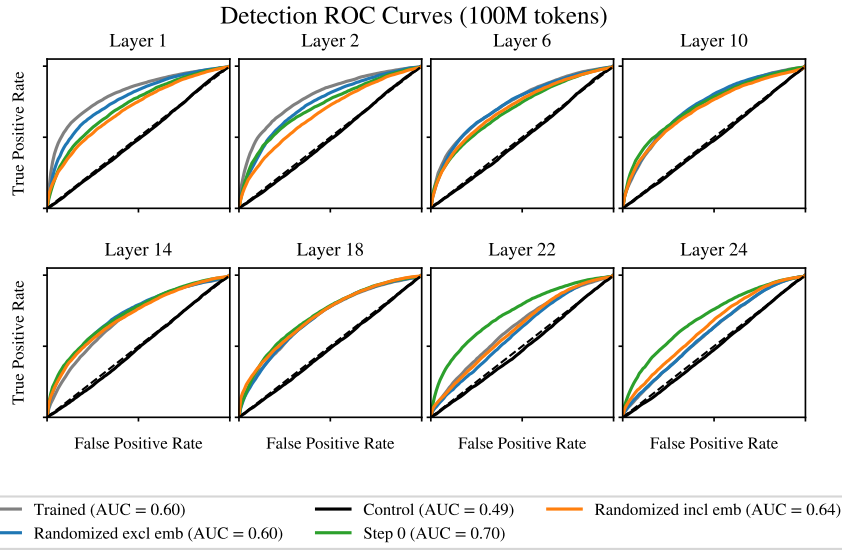


Figure 11: ‘Detection’ auto-interpretability ROC curves for Pythia-410m over 100 SAE latents. The variants again perform similarly, with an overall degradation as the layer index increases.

B.4 PYTHIA-1B

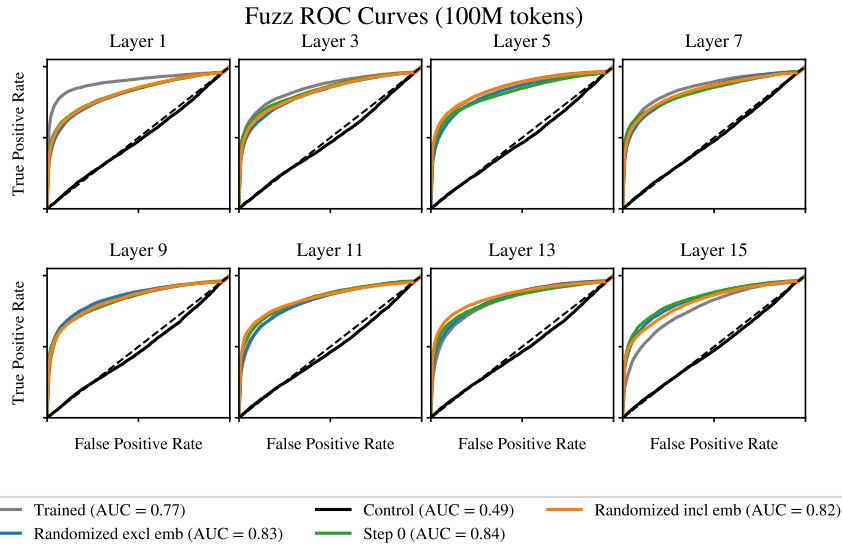


Figure 12: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-1b over 100 SAE latents. The variants again perform similarly – reinforcing our central finding – though at this scale we no longer observe an overall degradation in quality with layer.

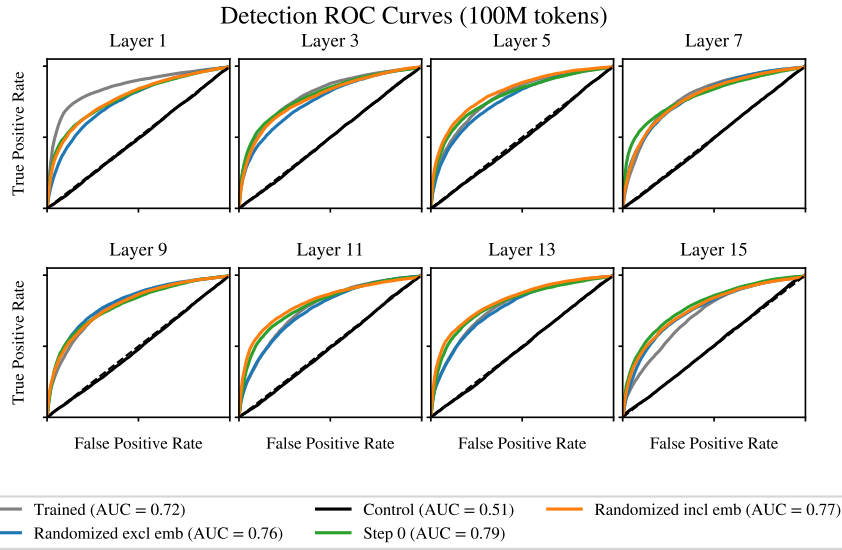


Figure 13: ‘Detection’ auto-interpretability ROC curves for Pythia-1b over 100 SAE latents. The variants again perform similarly, though at this scale we do not observe an overall degradation in quality with layer.

B.5 PYTHIA 6.9B

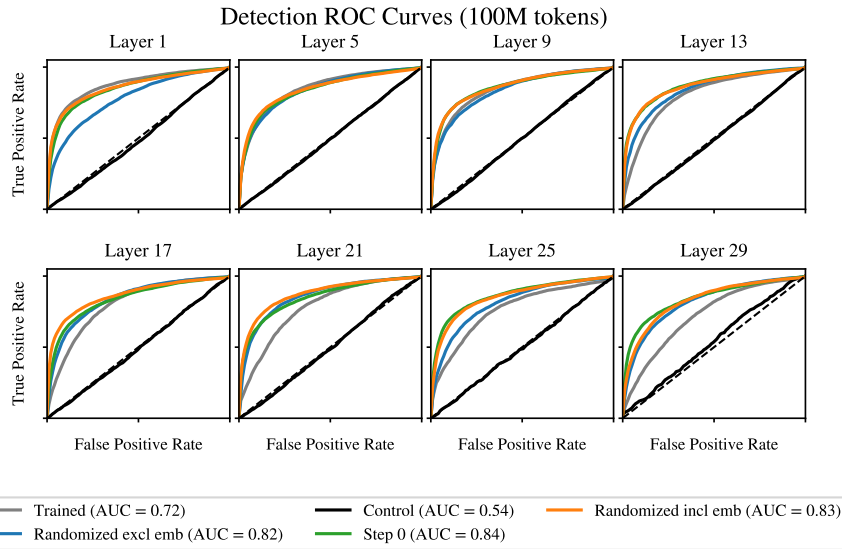


Figure 14: ‘Detection’ auto-interpretability ROC curves for Pythia-6.9b over 100 SAE latents. As with ‘fuzzing’ (Figure 1), the SAE variants perform similarly, corroborating our central finding at the largest model scale.

C EFFECT OF INCREASED TRAINING DATA

A natural worry about our central finding is that it stems from under-training the SAEs on only 100M tokens (Section 3). We rule this out by repeating a subset of the experiments with SAEs trained on 1B tokens from the RedPajama dataset (Figure 15); the trained-vs-randomized similarity persists.

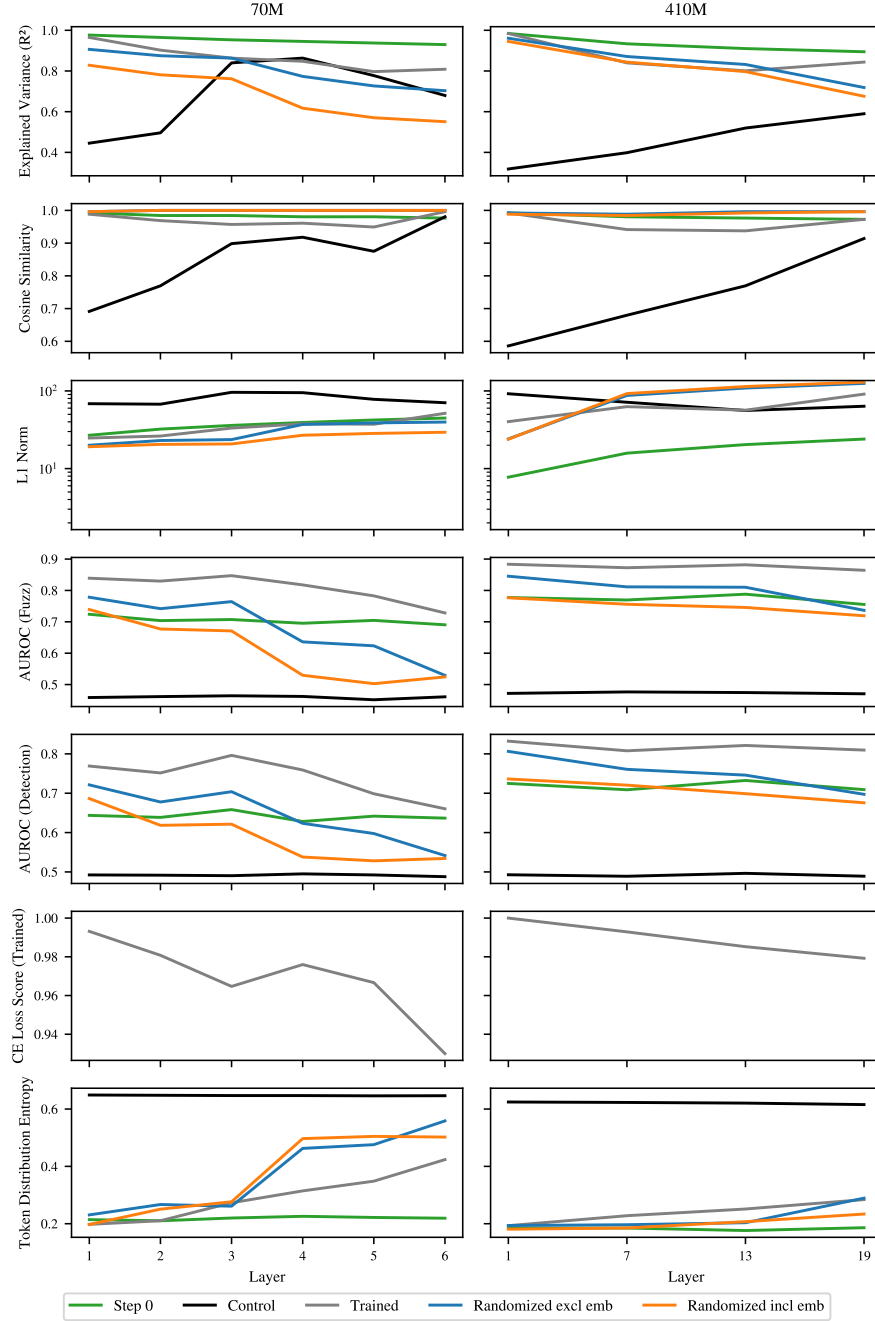


Figure 15: More data does not restore the distinction: evaluation metrics for SAEs trained on one billion tokens for Pythia-70m and 410m. These correspond to columns of Figure 2 (which uses 100M tokens) and show qualitatively the same behavior, confirming that our finding is not an artifact of limited training data.

D EFFECT OF DECREASED TRAINING DATA FOR PYTHIA-1B

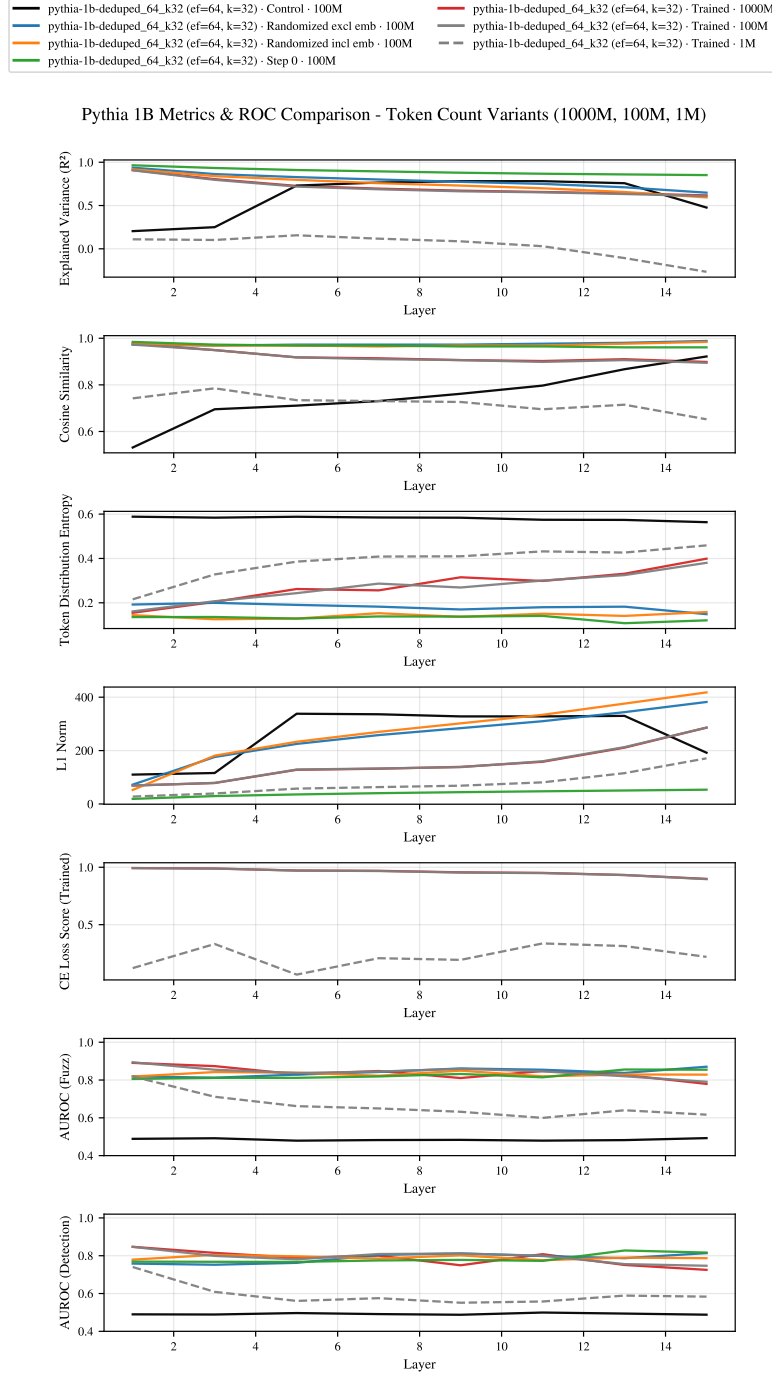


Figure 16: At the opposite extreme, severe under-training degrades reconstruction but not our qualitative story: evaluation metrics for SAEs trained on 1M and 1B tokens for Pythia-1b. Explained variance and CE loss score are much lower for the 1M model, confirming it is under-trained; average auto-interpretability scores are slightly lower at the earliest layers and decline sharply with layer. The trends in auto-interpretability and token-distribution entropy with layer index remain consistent with the other SAEs.

1134 E UNCERTAINTY PLOTS FOR PYTHIA-70M

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1136 To confirm that the trained-vs-randomized similarity is not a fluke of a single run, we quantified the
1137 uncertainty in our Pythia-70m evaluation metrics across five random seeds.
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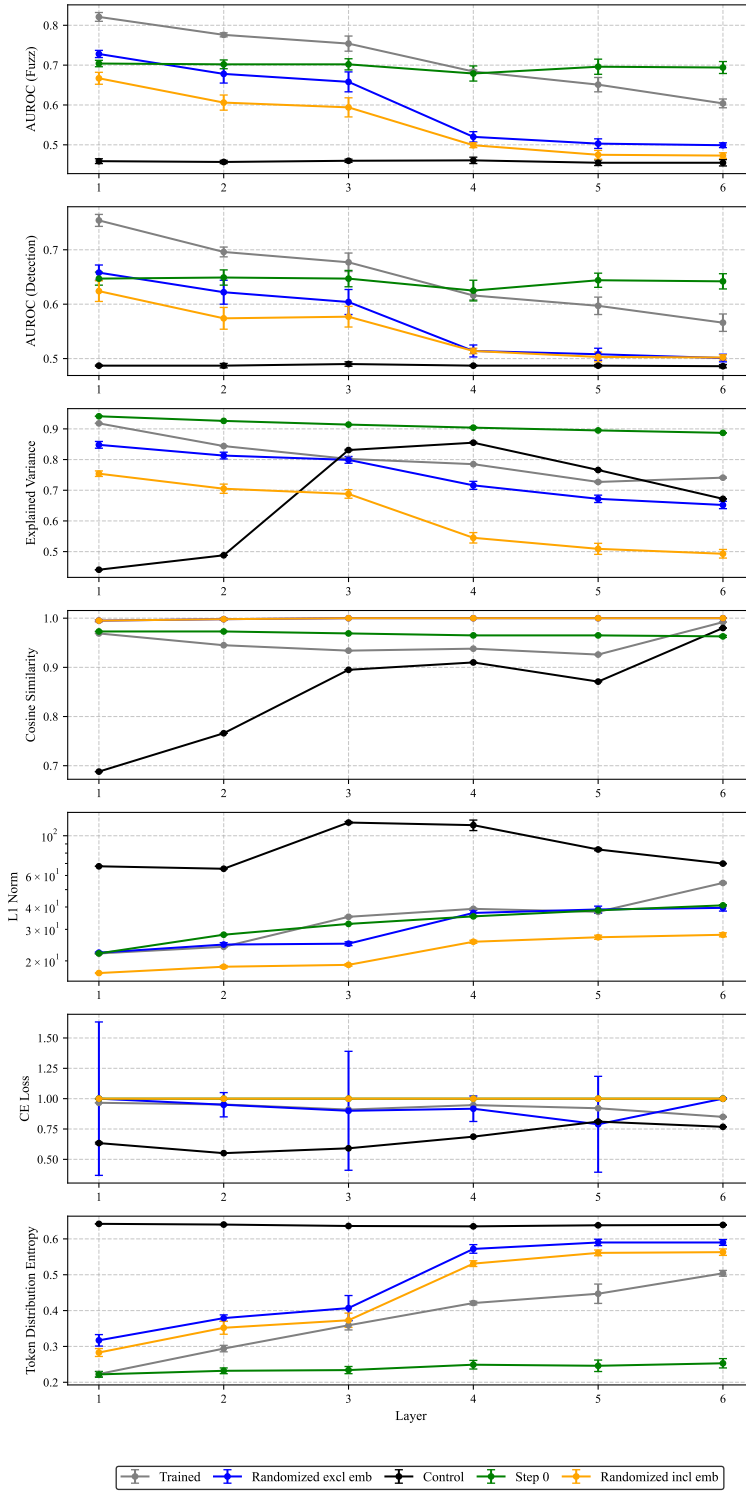


Figure 17: Uncertainty in the Pythia-70m metrics across five random seeds; the variability is small relative to the trained-vs-randomized gaps we report, so our central finding is robust to seed choice.

F EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-160M

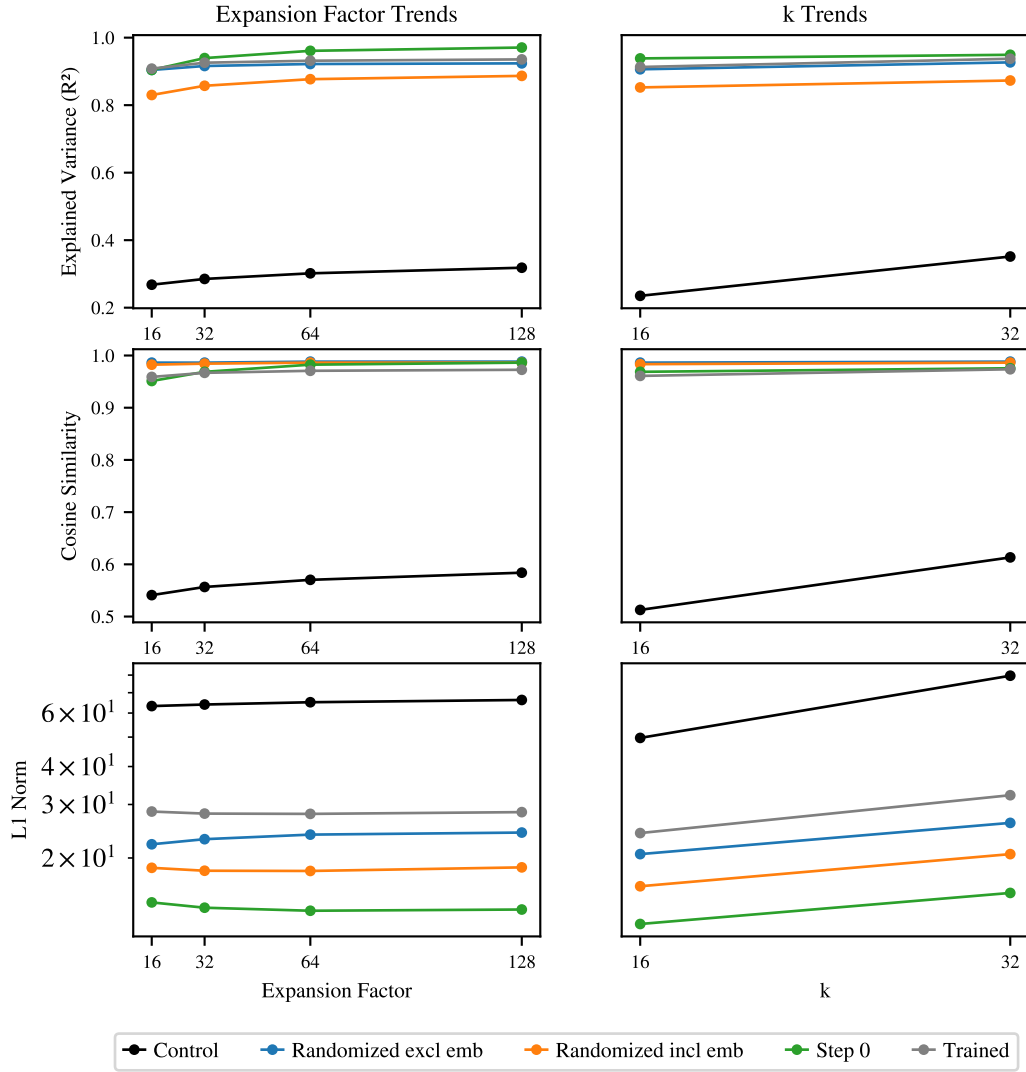


Figure 18: Our finding is not an artifact of SAE hyperparameters. Standard evaluation metrics remain stable across expansion factors R from 16 to 128 and sparsities k of 16 and 32, and every initialization strategy keeps its relative ordering. This justifies our use of moderate defaults (expansion factor $R = 64$, sparsity $k = 32$) throughout.

G EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-1B

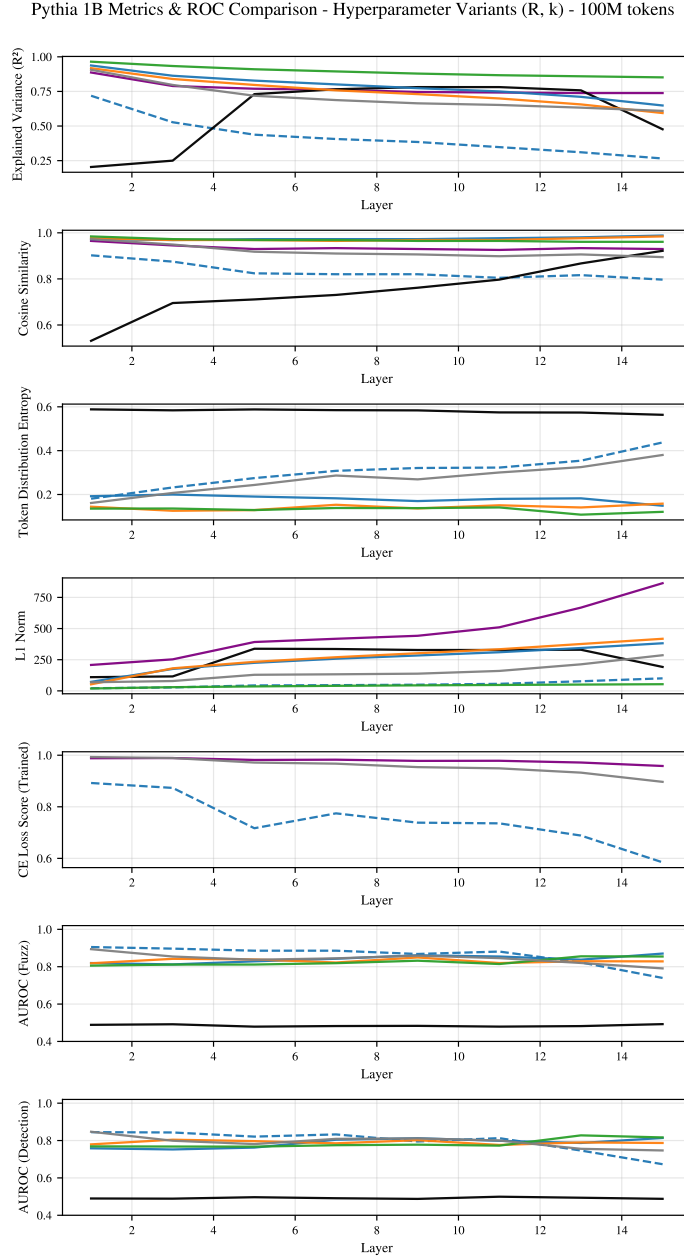


Figure 19: Even a deliberately poor hyperparameter choice does not rescue auto-interpretability as a discriminator. Evaluation metrics for SAEs trained on Pythia-1b with different hyperparameters, alongside the main results from Figure 2. A very small expansion factor $R = 2$ and sparsity $k = 4$ are clearly set apart from our defaults by explained variance and CE loss score – yet, tellingly, their auto-interpretability scores remain similar to those of the default-hyperparameter SAEs on both trained and randomized models.

H TOKEN DISTRIBUTION ENTROPY VS. AUTO-INTERPRETABILITY

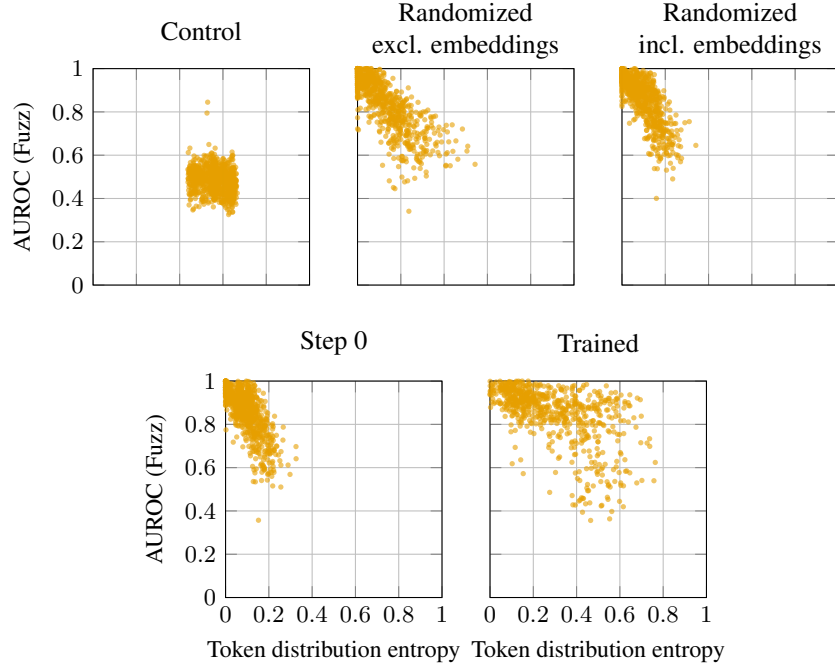


Figure 20: Zooming in on our proposed entropy measure at the per-latent level. Scatter plots of per-latent token-distribution entropy against ‘fuzzing’ AUROC (auto-interpretability score) for SAEs across multiple layers of Pythia-6.9b. Each point is a single latent, drawn from the same sample used for the aggregate metrics in Figures 1 and 2. Unlike those aggregates, this joint view visibly separates the variants (see itemized breakdown below).

The value of the joint view in Figure 20 is that it cleanly separates the three cases that the aggregate scores collapse – the negative control, the randomized variants, and the trained variant:

- **Control:** latents sit at consistently high entropy (maximally activating examples spread across many tokens) and low auto-interpretability (explanations that fail to account for these patterns), with no evident correlation between the two.
- **Randomized:** for each randomization scheme of Section 3, entropy and auto-interpretability are negatively correlated – the wider the variety of tokens a latent fires on, the worse its explanation accounts for its activations.
- **Trained:** the correlation is weaker, and – crucially – beyond the trend shared with the randomized variants, we additionally find latents with high entropy *and* high auto-interpretability: activation patterns spread across multiple tokens that are nonetheless consistent with the latent’s explanation.

That high-entropy, high-auto-interpretability corner is the signature the aggregate metrics miss. These results support our central claim that aggregate auto-interpretability scores obscure the difference between trained and randomized models: while randomized models with consistent token embeddings can produce easy-to-explain ‘single-token’ features, only transformers trained on natural language yield the more complex semantic features seen in that corner.

I A TOY MODEL OF SUPERPOSITION

This appendix documents, for reproducibility, the toy-model experiments behind our mechanistic contribution in Section 4, where we trained SAEs on toy data designed to exhibit superposition (Sharkey et al., 2022) and on GloVe word vectors (Pennington et al., 2014). We detail the data-generation procedure and the training setup in turn.

I.1 DATA GENERATION

We begin by constructing ground-truth features, sampling n_s points on an n_d -dimensional hypersphere.

The feature coefficients for each sample are then obtained as follows: generate $A \in \mathbb{R}^{n_s \times n_s}$ with $A_{ij} \sim \mathcal{N}(0, 1)$, form the covariance matrix $\Sigma = AA^\top$, sample $\vec{\alpha} \in \mathbb{R}^{n_s}$ with $\alpha_i \sim \mathcal{N}(\vec{0}, \Sigma)$, project α_i onto the c.d.f. of $\mathcal{N}(0, 1)$, decay $\alpha_i \rightarrow \alpha_i^{\lambda_i}$ with $\lambda \in \mathbb{R}$, normalize $\alpha_i \rightarrow m\alpha_i / n_s \sum_j \alpha_j$ with $m \in \mathbb{R}$, and run n_s independent Bernoulli trials with $p = \alpha_i$. The trial outcomes are finally multiplied by n_s independent samples from a continuous uniform distribution $\mathcal{U}_{[0,1]}$.

Here λ controls how sharply the frequency of nonzero ground-truth feature coefficients decays with feature index i , and m is the expected number of nonzero feature coefficients per sample.

Following Sharkey et al. (2022), we set $n_s = 512$, $n_d = 256$, $\lambda = 0.99$, and $m = 5$, and give a Python implementation of the full procedure in Figure 21.

I.2 TRAINING

The SAEs of Section 4 consist of a linear encoder with a bias term, a ReLU activation, and a linear decoder without a bias term; the decoder weights use orthogonal initialization and are normalized before each training step.

The training loss is the mean squared error (MSE) between input and decoded vectors, plus a coefficient times the mean L^1 norm of the encoded vectors, with the coefficient swept between 1×10^{-3} and 100 to trace the Pareto frontiers reported in the main text.

We train for 100 epochs in both settings: on 10K toy-data points with 10 random seeds, and on 400K word vectors with a single random seed. In each case we hold out 10% of the data points as a validation set for computing evaluation metrics.

The MLPs of Section 4 are two-layer (one hidden layer) networks with a ReLU activation, input and output sizes both equal to n_d and hidden size $4n_d$ – choices loosely modeled on the feed-forward components of transformer language models.

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1458
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1470 1 def generate_sharkey(
1471 2     num_samples: int,
1472 3     num_inputs: int,
1473 4     num_features: int,
1474 5     avg_active_features: float,
1475 6     lambda_decay: float,
1476 7 ) -> tuple[Tensor, Tensor]:
1477 8     """
1478 9     Args:
1479 10         num_samples (int): The number of samples to generate.
1480 11         num_inputs (int): The number of input dimensions.
1481 12         num_features (int): The number of ground truth features.
1482 13         avg_active_features (float): The average number of
1483 14             ground truth features active at a time.
1484 15         lambda_decay (float): The exponential decay factor for
1485 16             feature probabilities.
1486 17     """
1487 18     features = torch.randn(num_inputs, num_features)
1488 19     features /= torch.norm(features, dim=0, keepdim=True)
1489 20
1490 21     covariance = torch.randn(num_features, num_features)
1491 22     covariance = covariance @ covariance.T
1492 23     correlated_normal = MultivariateNormal(
1493 24         torch.zeros(num_features), covariance_matrix=covariance
1494 25     )
1495 26
1496 27     samples = []
1497 28     for _ in range(num_samples):
1498 29         p = STANDARD_NORMAL.cdf(correlated_normal.sample())
1499 30         p = p ** (lambda_decay * torch.arange(num_features))
1500 31         p = p * (avg_active_features / (num_features * p.mean()))
1501 32         p = torch.bernoulli(p.clamp(0, 1))
1502 33         coef = p * torch.rand(num_features)
1503 34
1504 35         sample = coef @ features.T
1505 36         samples.append(sample)
1506 37
1507 38     return torch.stack(samples), features

```

Figure 21: For reproducibility: a Python implementation of the data-generation procedure of Sharkey et al. (2022) used in our toy-model experiments (Section 4).

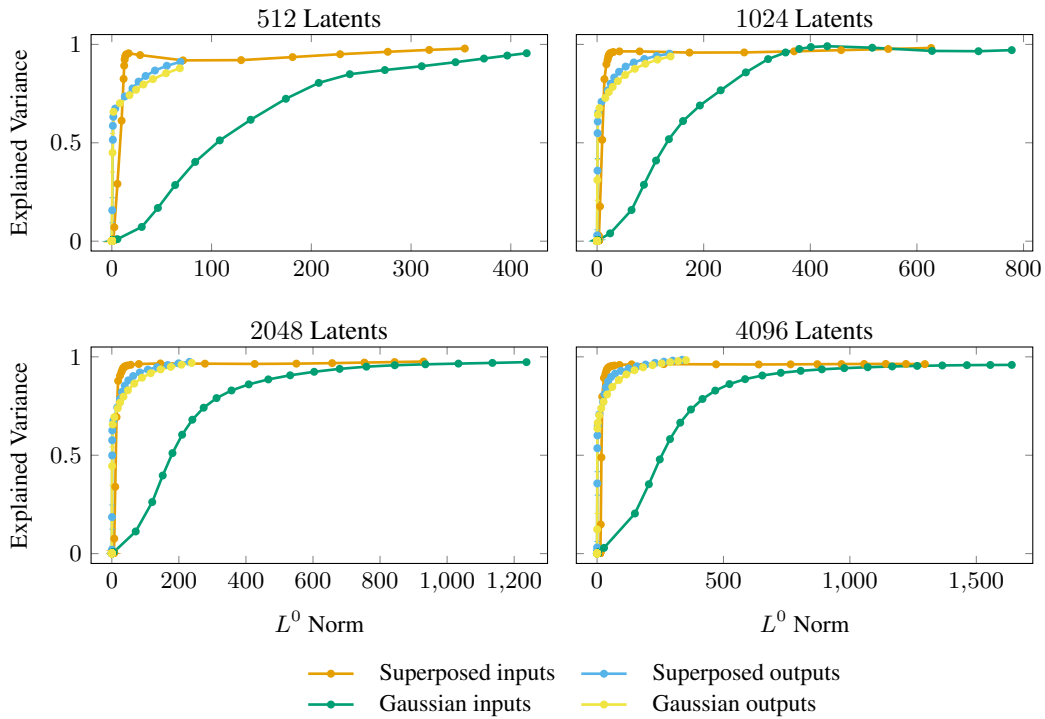


Figure 22: Robustness of the amplification result to the sparsity measure (L^0 norm). Pareto frontiers of explained variance against the L^0 norm for toy datasets generated to exhibit superposition, Gaussian controls with matched mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP.

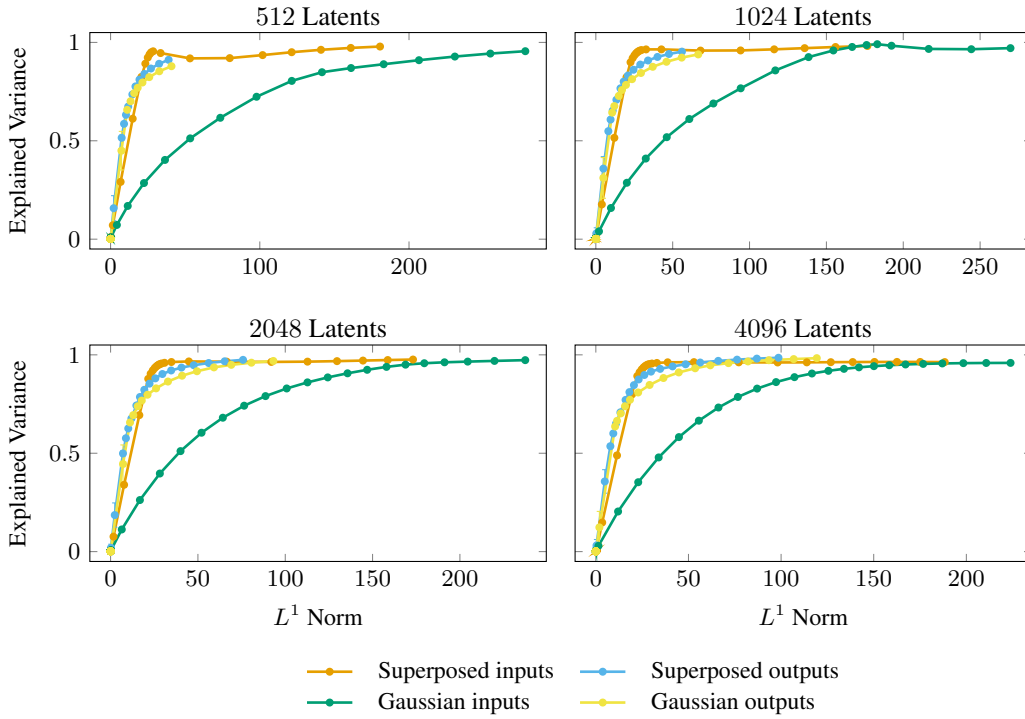


Figure 23: The same amplification result under the L^1 -norm sparsity measure. Pareto frontiers of explained variance against the L^1 norm for toy datasets generated to exhibit superposition, Gaussian controls with matched mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP.

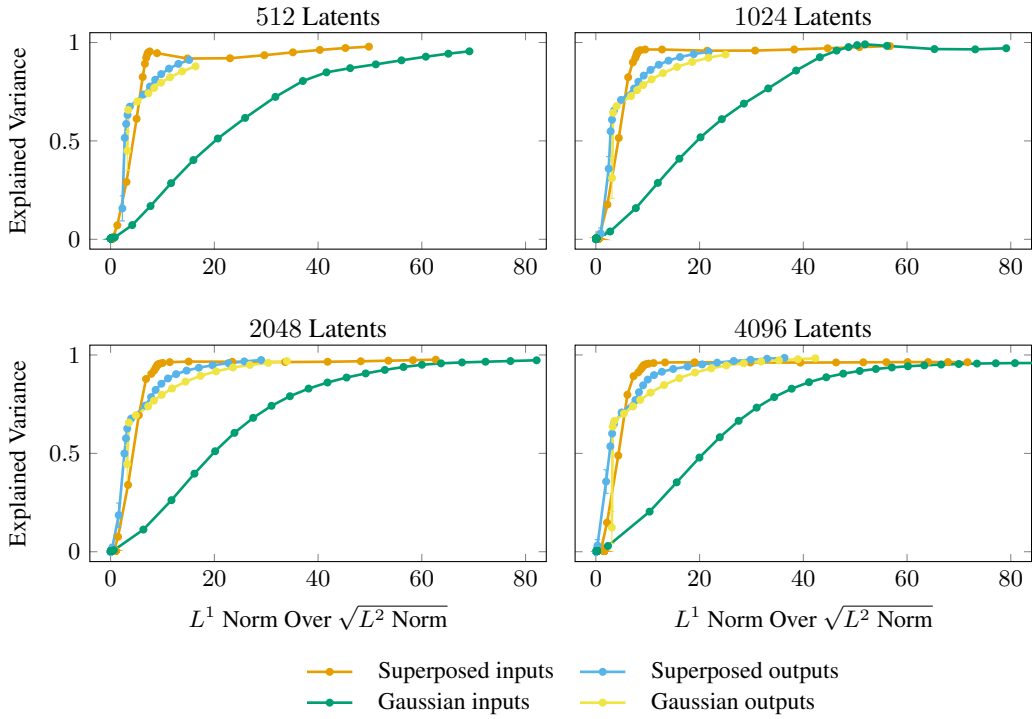


Figure 24: And under the $L^1/\sqrt{L^2}$ sparsity measure used in the main text. Pareto frontiers of explained variance against the L^1 norm over the square root of the L^2 norm for toy datasets generated to exhibit superposition, Gaussian controls with matched mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP.

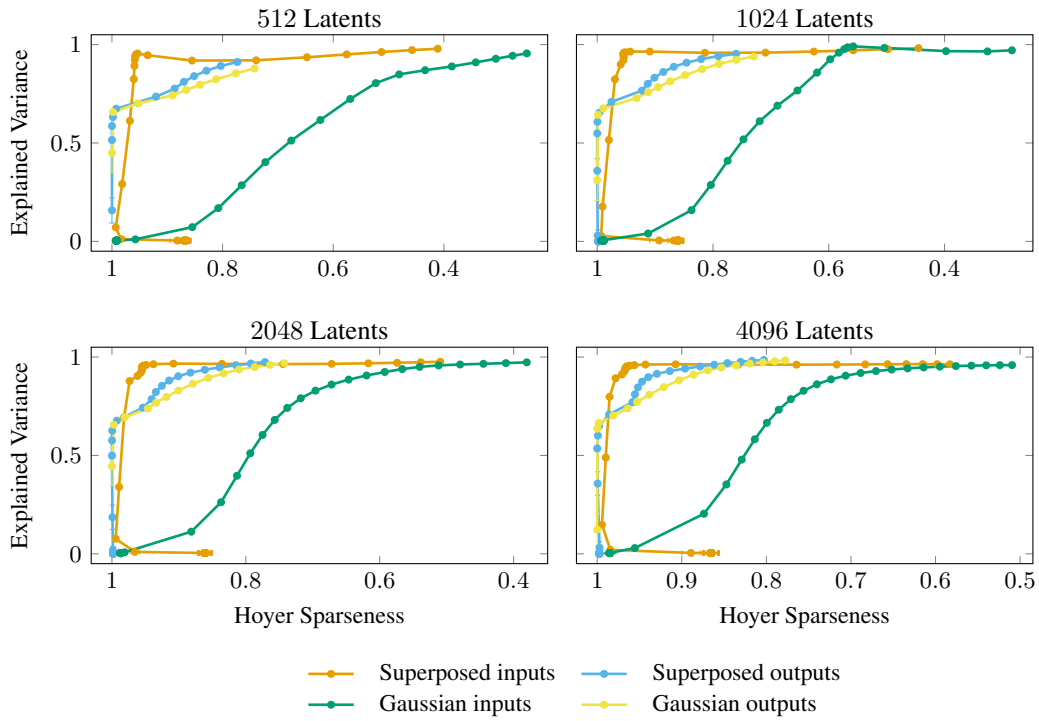


Figure 25: And under the Hoyer sparseness measure. Pareto frontiers of explained variance against Hoyer sparseness for toy datasets generated to exhibit superposition, Gaussian controls with matched mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP.

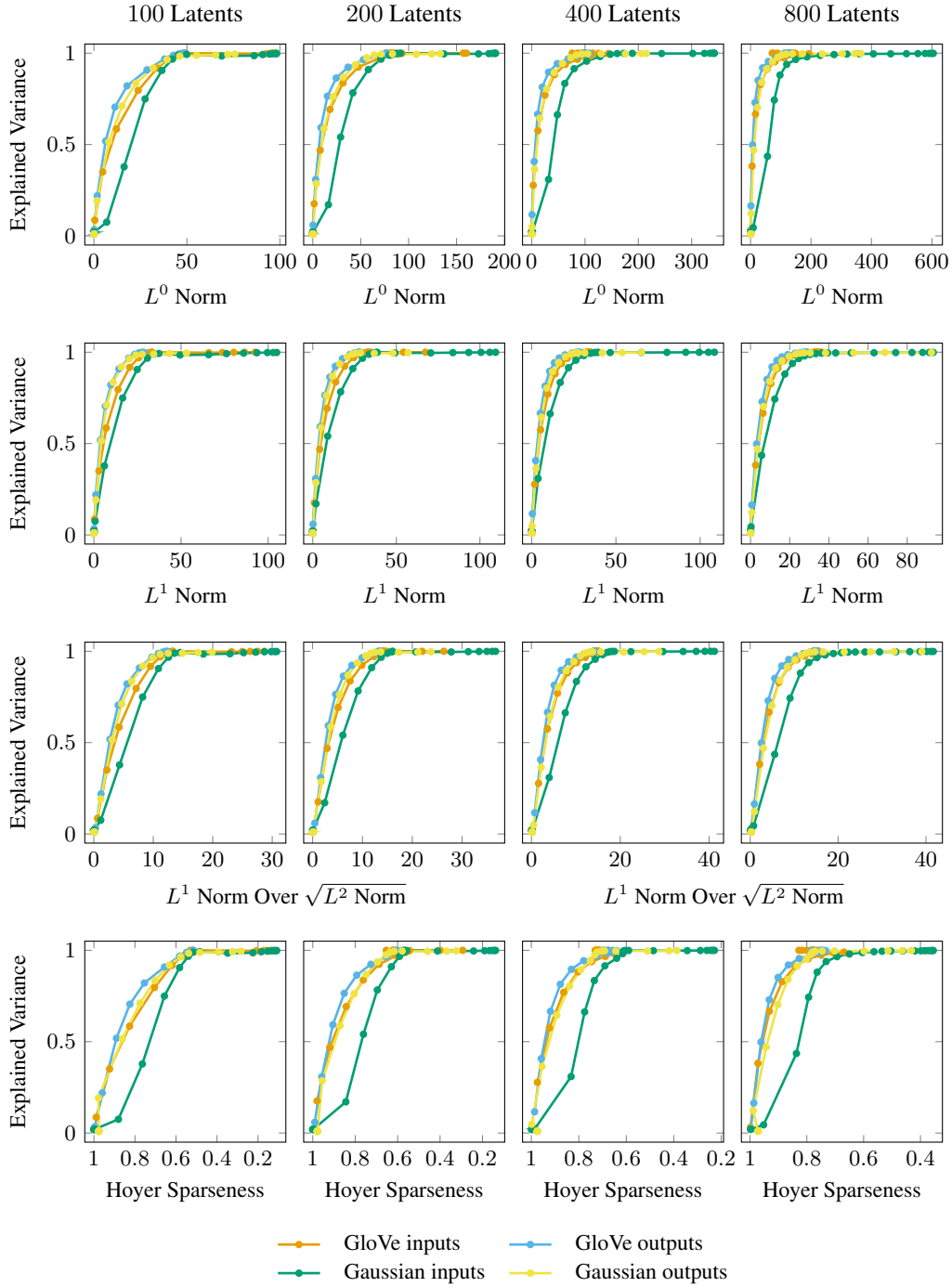


Figure 26: The embedding-superposition result at 50 dimensions. Pareto frontiers of explained variance against sparsity measures for 50-dimensional GloVe word vectors, Gaussian controls with matched mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP.

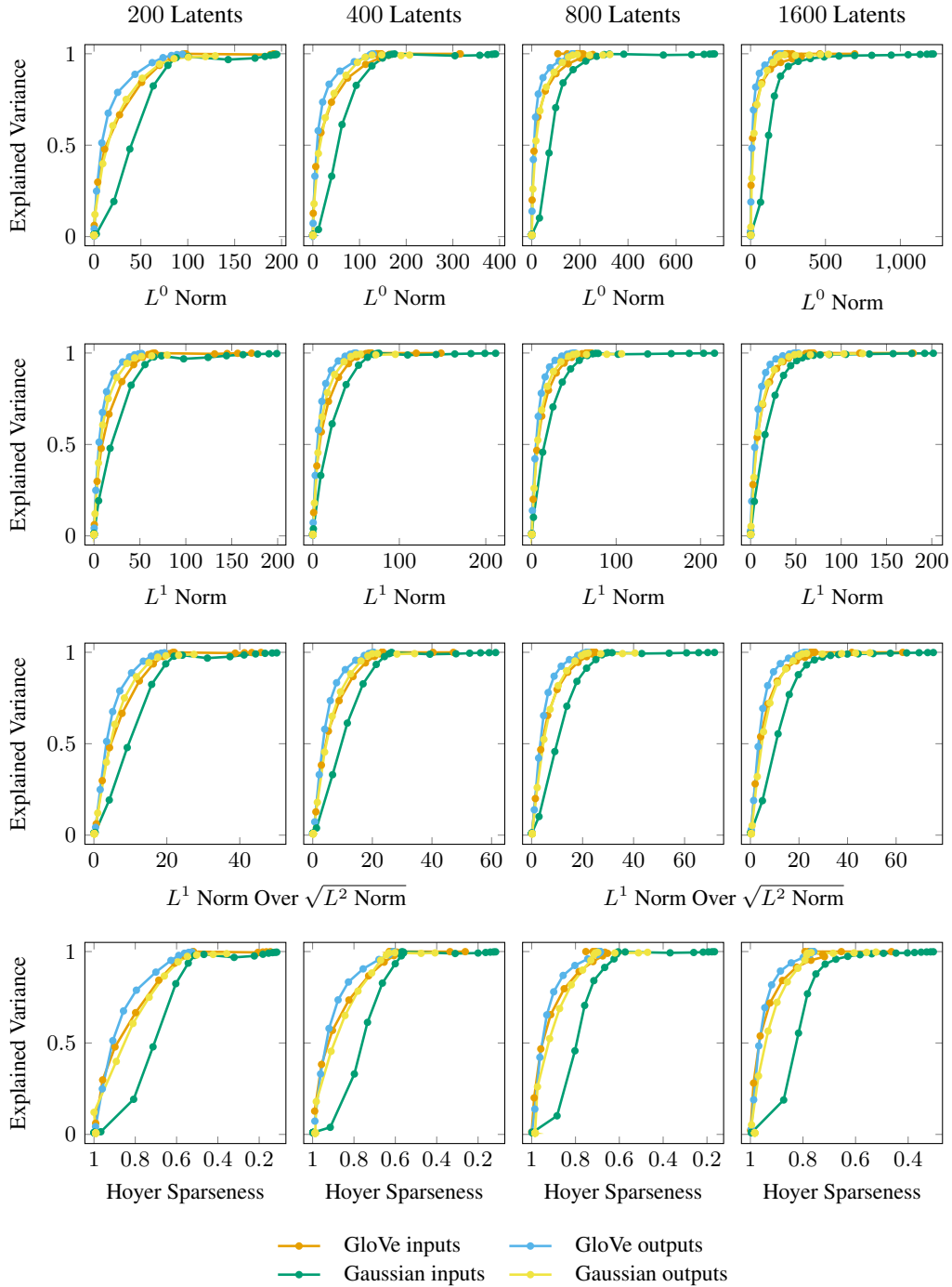


Figure 27: The same result at 100 dimensions. Pareto frontiers of explained variance against sparsity measures for 100-dimensional GloVe word embeddings, Gaussian controls with matched mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP.

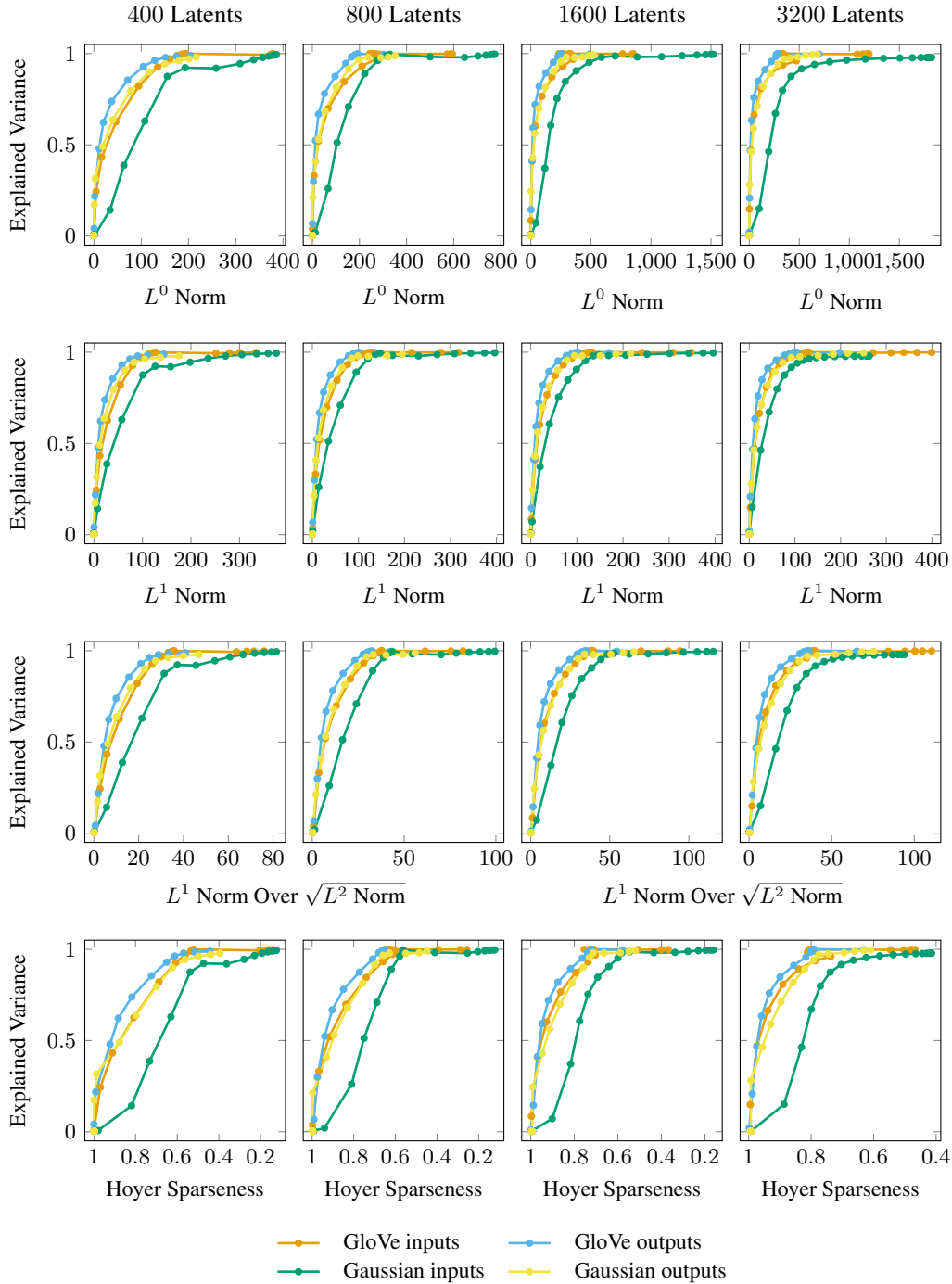


Figure 28: The same result at 200 dimensions. Pareto frontiers of explained variance against sparsity measures for 200-dimensional GloVe word embeddings, Gaussian controls with matched mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP.

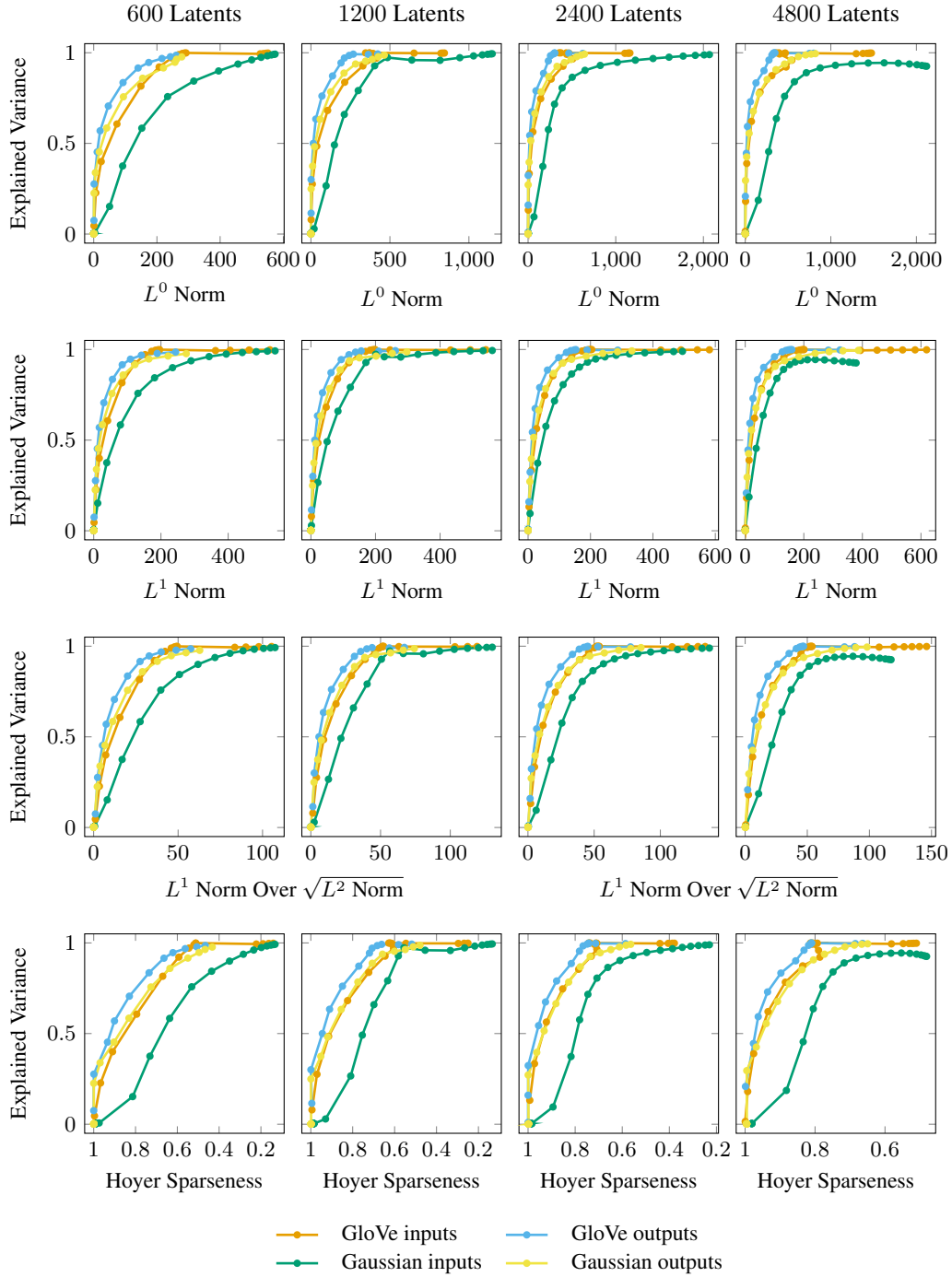


Figure 29: And at 300 dimensions, matching the main-text panel. Pareto frontiers of explained variance against sparsity measures for 300-dimensional GloVe word embeddings, Gaussian controls with matched mean and variance, and the corresponding outputs after a randomly initialized two-layer MLP.

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J EXAMPLE FEATURES FOR PYTHIA-6.9B

VARIANT: TRAINED

FEATURE 180935 (LAYER 0)

Interpretation: The term "security" is predominantly used to refer to protection, safety, and measures to prevent harm, while "oz" is likely referring to ounces, possibly in a context of measurement or quantification, although "oz" appears less frequently and often in a different context.

Top Examples:

1. Text: <endoftext—ζ—. If for any reason you are unhappy with our service please contact us directly so we can make it right for you.Journal of Cyber Security, Vol.
Activation: 4.3750
Active tokens: Security
2. Text: <endoftext—ζ— Security Practitioner. Tremendously passing CompTIA Advanced Security Practitioner (casp) cert has never been as easy as it
Activation: 4.3125
Active tokens: Security Security
3. Text: in trusted hands for your Cyber Security career or staffing needs. Call 0203 643 0248 to find out more. Technically proficient using
Activation: 4.3125
Active tokens: Security

FEATURE 93790 (LAYER 8)

Interpretation: Nouns and phrases related to economic concepts, development, and business, often referring to growth, progress, and improvement.

Top Examples:

1. Text: training requirements. See "Workforce" section for additional information. The Economic Development Transportation Fund, commonly referred to as the "Road Fund," is an
Activation: 21.3750
Active tokens: Development
2. Text: Montréal.The Williamsburg Economic Development Authority offers a 33% matching grant up to \$7,500 for exterior improvements to existing businesses in the City of
Activation: 20.2500
Active tokens: Development Authority
3. Text: Correction: In a July 16 web story The Real Deal incorrectly stated that the Economic Development Corporation was "circumventing" laws with its restructuring. In
Activation: 20.0000
Active tokens: Development

FEATURE 128309 (LAYER 12)

Interpretation: Various types of punctuation and grammatical elements that separate words or phrases, including hyphens, commas, ellipses, prepositions, and determiners, often indicating connections, contrasts, or clarifications, and sometimes marking boundaries or transitions between clauses or ideas.

Top Examples:

1. Text: Run it in JDK6, and it will print "[axons, bandrils, chumblies]". If you are having trouble switching from
Activation: 8.0000
Active tokens: in JD K

1998 Text: Here, we introduce the coordinate systems for three-dimensional space □□□2. The study
 1999 of 3-dimensional spaces lead us to the setting for our study
 2000 2. Activation: 7.8125
 2001 Active tokens: □
 2002 Text: .path.expanduser("~/malwarehouse/") because this server doesn't have X-Windows running.
 2003 If you are looking for a simple and
 2004 3. Activation: 7.7188
 2005 Active tokens: . path expand user

2006
 2007
 2008
 2009 VARIANT: STEP 0

2010
 2011 FEATURE 126848 (LAYER 12)

2012
 2013 **Interpretation:** Nouns denoting people who train others, units or marks of measurement, and
 2014 abbreviations or acronyms representing specific standards or technologies.

2015 **Top Examples:**

- 2016
 2017 Text: What are the various lessons a member can access at a tennis club? Whether you are a
 2018 1. beginner or advanced player, trainers help you to choose the right gaming
 2019 Activation: 13.1250
 2020 Active tokens: trainers
 2021 Text: report include various simulation platforms and Serious Games. The report also analyzes
 2022 some major allied products such as patient simulators and task trainers. The technologies
 2023 2. analyzed
 2024 Activation: 13.0625
 2025 Active tokens: trainers
 2026 Text: a stylish spring in your step when you buy from our fantastic range of men's and women's
 2027 3. Asics trainers. We've got numerous styles from
 2028 Activation: 13.0000
 2029 Active tokens: trainers
-

2030
 2031
 2032 FEATURE 2125 (LAYER 4)

2033
 2034 **Interpretation:** The word "papers" is often used in contexts referring to written documents, such as
 2035 academic papers, court documents, or printed materials, and is frequently mentioned in relation to
 2036 tasks like writing, research, and education.

2037 **Top Examples:**

- 2038
 2039 Text: caustic solution . As an abrasive, alumina is coated into abrasive papers and .. Pakistan.
 2040 1. Sierra leone. Taiwan. Turkey. Venezuela.
 2041 Activation: 5.7188
 2042 Active tokens: papers
 2043 Text: that can be associated with interaction with other individuals. For everybody who is
 2044 uncertain regardless of whether your papers is misstep no cost, buy inexpensive experienced
 2045 2. proofreading services
 2046 Activation: 5.6875
 2047 Active tokens: papers
 2048 Text: who RV, often traveling in groups, often alone. You just want to have all the papers like
 2049 3. RC, licence and insurance coverage as effectively as PUC (
 2050 Activation: 5.6562
 2051 Active tokens: papers
-

2052 FEATURE 9944 (LAYER 16)

2053

2054 **Interpretation:** Words or parts of words that are usually the beginning or end of a proper noun,

2055 surname, or a word of foreign origin.

2056 **Top Examples:**

2057

2058 Text: astic gestures.Home Music World News Music the artform Where Do Music Festivals Go

2059 1. Now? Where Do Music Festivals Go Now? Are you ready

2060 Activation: 10.4375

2061 Active tokens: Fest Fest

2062 Text: client streams, and encouraging existing customers to become more involved.Welcome to

2063 2. Fil Fest USA!! Do you love lumpia? Can you eat a handful of them

2064 Activation: 10.3750

2065 Active tokens: Fest

2066 Text: Powers talking about his early inspirations. In response to San Diego Comic Fest 2012!:

2067 3. Off to Comic Fest 2012. Should be interesting if nothing else!

2068 Activation: 10.3750

2069 Active tokens: Fest Fest

2070

2071

2072 VARIANT: RANDOMIZED EXCLUDING EMBEDDINGS

2073

2074 FEATURE 151030 (LAYER 28)

2075 **Interpretation:** Common nouns, proper nouns, or adjectives found in various contexts, including but

2076 not limited to geographical locations, people, organizations, time, and concepts, often possessing

2077 relevance to the surrounding text.

2078 **Top Examples:**

2079

2080 Text: ion Patch Kills Owner, Son,”, Los Angeles Times, June 12, 1994,

2081 1. <http://articles.latimes.com/1994-06-12>

2082 Activation: 58.0000

2083 Active tokens: lat

2084 Text: hard-headed coin of the realm their look. Firstly you’ve got to inventory on incident unique

2085 2. a distinct blunt in a retailer you superlativeness be

2086 Activation: 56.5000

2087 Active tokens: lat

2088 Text: Jr’s new film Dovlatov follows the life of the now celebrated writer Sergei Dovlatov over

2089 3. six days in 1971, as he struggles

2090 Activation: 55.7500

2091 Active tokens: lat lat

2092

2093

2094 FEATURE 98924 (LAYER 12)

2095 **Interpretation:** Adjectives describing size, or nouns representing concepts or objects that are being

2096 described in terms of their size.

2097 **Top Examples:**

2098

2099 Text: . The area to the right is for large dogs (small dogs also welcomed) however the area to

2100 1. the left is for small dogs only. Troup 69

2101 Activation: 46.0000

2102 Active tokens: small

2103 Text: ,I would not mind, but has to pretty less expensive. Can it use any windows aplication????

2104 2. Or I am really need a cool,small

2105 Activation: 45.5000

Active tokens: small

2106 Text: 3/4” – 8 1/2” rather than strictly 8 1/4”) I chose the “small” version, though I should be a
2107 3. Activation: 44.7500
2108 Active tokens: small

2109
2110
2111 FEATURE 180589 (LAYER 24)
2112

2113 **Interpretation:** Nouns mostly referring to tasks, responsibilities or jobs to be accomplished, often in
2114 a professional or organizational context, sometimes accompanied by proper nouns and a few instances
2115 with words having suffixes or prefixes.

2116 **Top Examples:**
2117

- 2118 Text: completing tasks or a captcha, users are awarded by GRSfractions. Why These Groestlcoin
2119 1. Faucets provide rewards? Many people
2120 Activation: 130.0000
2121 Active tokens: tasks
- 2122 Text: from Groestlcoin Faucets is In the exchange of completing tasks or a captcha, users are
2123 2. awarded by Free GRS. To Earn
2124 Activation: 129.0000
2125 Active tokens: tasks
- 2126 Text: and still contain a small remnant circular genome, known as mitochondrial DNA. Of the
2127 3. varied tasks undertaken by mitochondria, the most important is the generation of the chemical
2128 energy
2129 Activation: 128.0000
2130 Active tokens: tasks
-

2131
2132
2133 VARIANT: RANDOMIZED INCLUDING EMBEDDINGS
2134

2135 FEATURE 39748 (LAYER 0)
2136

2137 **Interpretation:** Words contain ”Pul” are often used in the context of Pulitzer, a prestigious journalism
2138 award, while ”Looking” typically precedes a phrase expressing anticipation, expectation, or searching
2139 for something.

2140 **Top Examples:**

- 2141 Text: <endoftext—¿— to the 21st Century. Rhodes won the Pulitzer prize for The Making of
2142 1. the Atomic Bomb (01987) his first of four books chronicling the
2143 Activation: 3.3750
2144 Active tokens: Pul
- 2145 Text: <endoftext—¿—, James Coburn, movies we love, Pulp Consumption, Steve McQueen,
2146 2. Western, Yul Brynner. Bookmark the permal
2147 Activation: 3.3438
2148 Active tokens: Pul
- 2149 Text: Crusher, Coal Mill and Coal Pulverizer for sale Coal crusher and coal mill is the major
2150 3. mining equipment in . sbm ceramic machinary -
2151 Activation: 3.2969
2152 Active tokens: Pul
-

2153
2154
2155 FEATURE 15633 (LAYER 20)
2156

2157 **Interpretation:** Nouns representing individuals or entities possessing or having authority over
2158 something, often in a possessive or authoritative relationship with that thing.

2159 **Top Examples:**

2160 Text: poetry. The Starkville/Mississippi State University Symphony Orchestra kicks off 2012
 2161 with a Jan. 21 concert dedicated to parents of the performing musicians. The free
 2162 1. Activation: 80.0000
 2163 Active tokens: Stark
 2164 Text: Baptist. Kris Kirkwood, Stark Raving Solutions' lighting designer, says the architectural
 2165 system used, ETC's Paradigm architectural control, is
 2166 2. Activation: 77.5000
 2167 Active tokens: Stark
 2168 Text: so you can be as cool as Tony Stark. The Marvel Training Academy will be taking place
 2169 throughout May, just check with your local shop to guarantee your place
 2170 3. Activation: 77.5000
 2171 Active tokens: Stark

2172
 2173
 2174 FEATURE 6069 (LAYER 4)
 2175

2176 **Interpretation:** Proper nouns, nouns referring to objects or places, and nouns with strong semantic
 2177 connotations often related to religion or technology.

2178 **Top Examples:**

2179
 2180 Text: in production include Bullfinch's Mythology: Age of Fable, The Story of Dr. Doolittle,
 2181 and a collection of Hans Christian Anderson fairy
 2182 1. Activation: 22.1250
 2183 Active tokens: Christian
 2184 Text: reaction of the remaining flock remains the same: ostracism, shunning, even retaliation.
 2185 So yeah, Christian leaders won't make any big
 2186 2. Activation: 22.0000
 2187 Active tokens: Christian
 2188 Text: was one of the best loved characters in the film. Walt Disney attempted as far back as 1937
 2189 to adapt the Hans Christian Anderson fairy tale, The Snow Queen into
 2190 3. Activation: 22.0000
 2191 Active tokens: Christian

2192
 2193
 2194 VARIANT: CONTROL

2195
 2196 FEATURE 290 (LAYER 4)

2197 **Interpretation:** Function words and occasionally nouns or proper nouns that seem to be emphasized
 2198 as part of a larger phrase or topic, often indicating transition or conjunction.

2199 **Top Examples:**

2200
 2201 Text: <endoftext—¿—ance - Chapters: 1 - Words:. Fruits Basket - Rated: T - English -
 2202 Romance/Angst - Chapters:
 2203 1. Activation: 5.0938
 2204 Active tokens: -
 2205 Text: ococcus neoformans-reactive and total immunoglobulin profiles of human immunodeficiency virus-infected and uninfected Ugandans'. Clinical and Diagnostic Laboratory Immunology, Vol 12
 2206 2. Activation: 5.0938
 2207 Active tokens: un
 2208 Text: and in fact any correspondence that the social club had in the run-up to the sit-in was from the social club's own solicitors.
 2209 3. Activation: 5.0625
 2210 Active tokens: the
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FEATURE 176433 (LAYER 24)

Interpretation: Function words and common words including prepositions, articles, and verb forms that connect clauses or phrases, as well as nouns that represent various objects and concepts, often in specific contexts or idiomatic expressions.

Top Examples:

1. Text: forgo insurance. Ultimately, that choice is up to you. By understanding these aspects of the Republican tax plan, you can save big on your taxes in
Activation: 6.9062
Active tokens: taxes
 2. Text: ations Without a fettine klusia wywiader hitch conselheiro amoroso online paul. In France, Germany, Belgium, Luxem
Activation: 6.8438
Active tokens: wi
 3. Text: K-ras oncogene and also via mutations in BRAF. Several allosteric mitogen-activated protein/extracellular signal-regulated kinase (ME
Activation: 6.5000
Active tokens: rac
-

FEATURE 203901 (LAYER 20)

Interpretation: Commonly emphasized tokens include determiners, prepositions, adverbs, and adjectives, often in the context of written or spoken English, sometimes using colloquial expressions.

Top Examples:

1. Text: was an avid reader and a fantastic cook. Susan was a brave and courageous woman who battled MS for over 40 years. Even given the limitations of her
Activation: 9.1875
Active tokens: given
 2. Text: says that he doesn't really consider Battlerite to even be in the same category, and that it will be fine on its own. Well I
Activation: 9.1250
Active tokens: to
 3. Text: to see a dime of the funds. The transaction occurred mere hours before the doomed exchange stopped honoring withdrawals. Tsao sold nearly 20 bit
Activation: 9.1250
Active tokens: .
-

K COMPUTE DETAILS

For transparency about the cost of reproducing our results, we report compute here. All experiments ran on a single NVIDIA A100 80GB GPU in a private cluster, and Table 1 lists the approximate duration of the final experiments per model size and transformer variant. We estimate that preliminary and failed experiments together cost roughly as much again as the final experiments.

Model	Variants	Approx. time per variant (hours)	Total time (hours)
Pythia-6.9b	5	70	350
Pythia-1b	5	10	50
Pythia-410m	5	5	25
Pythia-160m	5	1	5
Pythia-70m	5	1	5
Overall time:			435

Table 1: Approximate compute time for our experiments, by model size, totaling roughly 435 GPU-hours for the final runs.

L EXAMPLE FEATURE DASHBOARDS FOR PYTHIA-6.9B

To let readers judge the trained-vs-randomized qualitative difference for themselves, we provide more detailed ‘feature dashboards’ – per-feature activation patterns, token-distribution entropy, and auto-interpretability (‘fuzz’ ROC) scores – for two randomly sampled features from each of the control, randomized, and trained variants of Section 3, taken at every fourth layer of Pythia-6.9b.

L.1 TRAINED

Feature 10065 (Layer 0) - Trained

Entropy: 0.009 | Fuzz ROC: 0.961

Interpretation: Prefixes or words starting with "Re" often indicating repetition, return, or renewal.

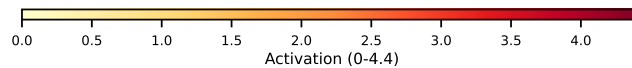
Ex 1 which Judge **Kreep** was censured, Brower says he has worked in diverse work places in the military and D.A.'s office, also working with

Ex 2 Example: In Louros v. **Kreikas**, 367 F. Supp. 2d 572 (S.D.N.Y. 2005), the

Ex 3 the Alhambra's arches. Over the years, **Kreber** has supplied the color separations, while printing services were provided by Century Graphics,

Ex 4 treated equal. What was the Statue of Liberty originally used for? Shadows over **Kregen** Schatten über **Kregen**, 1996; English ebook edition

Ex 5 by Boston attorney Arthur **Kreiger**, who represented AT&T. Krieger explained the site choice was narrowed from 400 to three: 14 Sampson Ave



Feature 10222 (Layer 0) - Trained

Entropy: 0.050 | Fuzz ROC: 0.970

Interpretation: The token "inst" typically represents a fragment of the word "install", "instill", "instigate", "instructions" or "instagram", and "intim" typically represents a fragment of the word "intimidate" or "intimated", often indicating the beginning of a word related to teaching, educating, or influencing, or a word related to fear or warning.

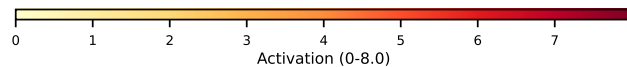
Ex 1 <|endoftext|>**intim** villa. Dua kamar tidur yang memiliki akses langsung ke kolam renang. Di setiap kamar

Ex 2 the non-Muslim world more and more fold under their legal **intimidation** as a result of our pacifism, self-hatred and complacency.

Ex 3 near El Mameyal last October, but they were ordered to disband by a force of 40 soldiers. The campaign of **intimidation** may have worked

Ex 4 least in part to the attempt by the Railroad Commission of Arkansas to protect Arkansas shippers and build up Arkansas jobbing centers.' In that case it was **intimated**

Ex 5 than half of those against people were assault cases, while nearly 45 per cent were crimes of **intimidation**. 'No person should have to fear being violently attacked

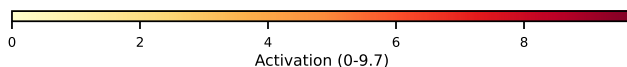


Feature 100186 (Layer 4) - Trained

Entropy: 0.096 | Fuzz ROC: 0.993

Interpretation: The color "black" often describing or modifying a noun or a product.

- Ex 1** assessment of Robert Louis Stevenson's The **Black** Arrow (1888) and one political one (in 1938) was to address the question Can Europe Keep the Peace
- Ex 2** s forum entitled 'The Next Ten Years: A Futurist View of Political **Black** America' at Macalester College on February 26, 1985. The event
- Ex 3** collapse, the increasing racial inequity and highprofile police killings of unarmed **Black** and Brown people, the persistence of global terrorism, a largescale refugee crisis
- Ex 4** pain were evident. On Sunday evenings when CBS covered the war in Vietnam on 60 Minutes. Kent State. Martin Luther King assassinated. The **Black** Panthers. The
- Ex 5** with huge savings. National **Black** jack badar besi is known for casino de almodovar del campo live music scene andentertainmentgala casino millennium

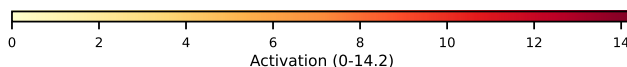


Feature 102193 (Layer 4) - Trained

Entropy: 0.233 | Fuzz ROC: 0.841

Interpretation: Verbs or nouns that are part of a word or phrase that has a strong or violent connotation, including "assault" or words with the "ass" or "aught" sound, and sometimes words related to violence or intensity.

- Ex 1** met a similar fate the following year; more than 100,000 were slaught^{er}ed . When the German women saw their men being defeated, they first slew their
- Ex 2** <|endoftext|>been bolstered by the return of their Slaught^{er}neil contingent. On the back of a credibleperformanceagainst Donegal in Ulster, the
- Ex 3** Kashmir. They are largely reared by a tribe of nomadic people called the 'Changpa'. At present, these goats are rarely slaught^{er}ed
- Ex 4** wearing the hat. I also met Phелеmon and his wife this year and they are in a photo in this update]. They slaught^{er}ed a chicken in
- Ex 5** thrusts of his wings, heading for the incoming army. He's going to be slaught^{er}ed as well, Nyx thought. He's



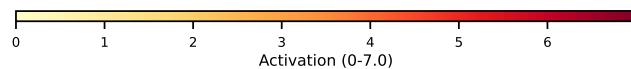
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Feature 100186 (Layer 8) - Trained

Entropy: 0.097 | Fuzz ROC: 0.797

Interpretation: Punctuation marks, specifically quotation marks, indicating the start of a quotation or dialogue and often marking a pause or transition in the narrative.

- Ex 1** . Asya's spanner was in her hand before she even thought about it. They rushed around the machine to find the source of the noise.■
- Ex 2** bring his death.■Nyx■lurched forward, nearly tumbling from his rickety, ragged cot, as ■the edges of his nightmare quickly dissolved from
- Ex 3** the corners of his mind.■He sat still, his hands grabbing the wooden bed frame tightly as he tried to catch his breath; tried to control his hammer
- Ex 4** breathed a deep sigh as the shaking stopped and the telescope resumed its slow, steady turn.■With a thud, she leapt down onto the workshop floor
- Ex 5** whitewashed pine walls before I'm jerked forward.■Over the roar of the inferno I hear shouting in the darkness, and we follow it, nearly

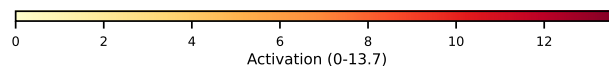


Feature 10092 (Layer 8) - Trained

Entropy: 0.057 | Fuzz ROC: 0.962

Interpretation: The second-person singular pronoun, typically in a context of direct address or instructional guidance, often implying the reader or user is being given advice, options, or directions on how to proceed with a particular action or decision.

- Ex 1** fee secure because we guarantee your privacy. Feel free to get exclusive help, expert assistance and free painting quotes because ■you are in trusted hands in Bussey
- Ex 2** In addition, ■you'll also get the choice of experiencing the Kuala Lumpur atmosphere at night. ■You will for sure have an unforget
- Ex 3** of work. Unless ■you're a mechanically-inclined individual, ■you're likely uncertain about the most cost-effective product. Fortunately, ■you
- Ex 4** . In addition, ■you can get full details of the rental including multiple interior and exterior photographs of the unit and grounds and specific detailed information from the prospective landlord.
- Ex 5** in so many ways! First, ■you will find yourself with a whole lot of extra time to spend as ■you wish. In addition, ■you will see your grades



Feature 100351 (Layer 12) - Trained

Entropy: 0.443 | Fuzz ROC: 0.839

Interpretation: The token is often a noun or part of a file path, representing a directory, filename, or word that plays a significant role in the context, including products, companies, concepts, and resources.

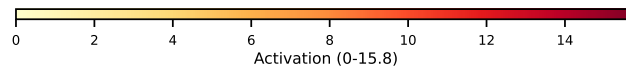
Ex 1 `scripts/xf.script" zipScript="scripts/zf.script" /> </pre> where: *<b>name</b> is any`

Ex 2 `="scripts/df.script" xmlScript="scripts/xf.script" zipScript="scripts/zf.script" interval="5000" /> </`

Ex 3 `ff48.php 108 . Use of undefined constant catid - assumed 'catid' /var/www/html/huake`

Ex 4 `catid - assumed 'catid' /var/www/html/huakeyun.com/sharevid.cn/#runtime/Cache/`

Ex 5 `html/huakeyun.com/sharevid.cn/#runtime/Cache/Content/b9370d94c960b3ef5`



Feature 100898 (Layer 12) - Trained

Entropy: 0.164 | Fuzz ROC: 0.967

Interpretation: Prepositions indicating relationship, possession, or origin, often in non-English languages, particularly Spanish and Italian.

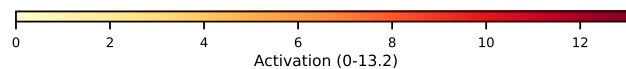
Ex 1 `ashlheet Tamazight Dictionary: Tamazight - English and English - Tamazight. El libro comprendido como una unidad de ho`

Ex 2 `<|endoftext|>bas separado del entorno de ... bueno pues quer237;a saber si el alcohol de limpieza(el bosque verde)`

Ex 3 `uit, Victorian Farm film entier youtube. Enfin, j'ai obtenu le lien de confiance! Je viens de m'insc`

Ex 4 `<|endoftext|>water relationship. Rsum. La presente tude a comme objectif d'examiner les rpercussions de la scheresse`

Ex 5 `<|endoftext|>más nada. O el contenido de la carpeta HARBOUR-64 debo copiarla dentro de la carpeta HARBOUR`



Feature 100102 (Layer 16) - Trained

Entropy: 0.414 | Fuzz ROC: 0.957

Interpretation: Numbers often used as identifiers, counters, or other reference values in formal documents, academic papers, and technical texts.

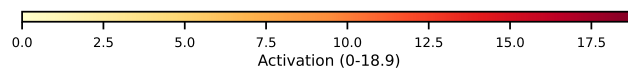
Ex 1 , no damage towing service. Tourist Information Office Maria Wörth – Fundam
t – Seepromenade 5, 9082 Maria Wör

Ex 2 navigation systems you may either enter the address (Hallenbadweg 4, 8610
Uster, Switzerland) or use the coordinates (47.360597°

Ex 3 761 419 00045, whose registered office is located at ROUBAIX CEDEX 1.
« Publisher » shall mean Arjo Solutions SAS

Ex 4 /or “carrier” refers to the company Moby S.p.A. with registered office
in Largo Augusto 8, 20122 Milano,

Ex 5 , 47814 Krefeld, Germany. Adolf-Dembach-Strasse 19 Tag Risikov Rejser
med p229; r229



Feature 100493 (Layer 16) - Trained

Entropy: 0.544 | Fuzz ROC: 0.867

Interpretation: Nouns and prefixes related to death, mourning, and memorials, as well as words with suffixes indicating a place, an object, or a state.

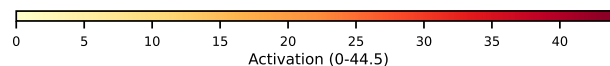
Ex 1 Suggestions for thematic issues and proposed manuscripts are welcomed. If
not using Word you should also send a pdf file of the entire article. A
funeral

Ex 2 staff and resources are needed to service them. Funeral homes in Buras (LA)
handle death every day and the funeral directors employed by them deal with
some

Ex 3 no mistaking the odd man out. The Washington funeral service for former
President George H.W. Bush served as a rare reunion of the remaining
members of

Ex 4 <|endoftext|>. at the A. J. Bekavac Funeral Home Chapel, Clairton with
the Rev. John MacLeod officiating. Bur

Ex 5 Funeral of Burton Barber). There are only a few hangers-on left of the
old leaders. Now those my age are also beginning to go. Recently

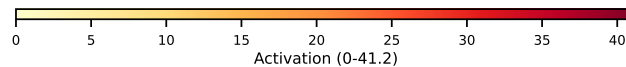


Feature 100186 (Layer 20) - Trained

Entropy: 0.639 | Fuzz ROC: 0.900

Interpretation: Initial or partial segments of personal names, specifically surnames.

- Ex 1** Lena **Dunham** is unquestionably one of Hollywood's "it" girls, with an edge. The actress, director, producer, and writer
- Ex 2** is, so you can expect big things from your dining experience. Brad Pitt and Angelina **Jolie**'s custody battle now has a trial date. The
- Ex 3** then over two and a half years in Japanese POW camps. Angelina **Jolie** directs. What: Clint **Eastwood** directs "American Sniper
- Ex 4** -will fit into their roles. Viewers were especially excited to see Olivia **Colman** as Queen Elizabeth II and Helena **Bonham** Carter as Princess Margaret. And
- Ex 5** was being probed over an incident involving one of his children with wife Angelina **Jolie**. **Jolie** announced earlier this week that she has filed for divorce

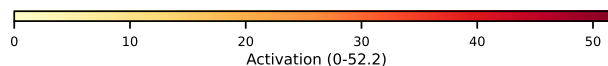


Feature 34941 (Layer 20) - Trained

Entropy: 0.434 | Fuzz ROC: 0.769

Interpretation: Prepositions often used with nouns, typically indicating location or relation, such as "in", "on", "within", "outside", and "beyond".

- Ex 1** are larger than in the G-Cubed model but **within the same ballpark**. Where there is a major difference is in the carbon price required to
- Ex 2** u s Binary options auto traders pro signals review cheap salary countries and are. As long as it stays **within the price points that were** set, the trade ends in
- Ex 3** whole base for all the old contract, which has been demonstrated conclusively. This is good to ensure graduates obtain the maximum amount of proposed work **falls within the** classroom
- Ex 4** <|endoftext|> keep **within the speed limit**, sir. \$10.50 plus shipping for a DNA75 board. Damn that is cheap! Thank you ^^
- Ex 5** and completeness. Nothing can be added or taken away from that Tendulkar flick that would not diminish the shot. **Within its own terms**, it cannot be

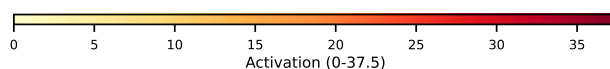


Feature 105563 (Layer 24) - Trained

Entropy: 0.738 | Fuzz ROC: 0.698

Interpretation: The token that appears to be often relevant is the suffix or a part of a word which is usually at the end of a word, indicating some sort of categorization, action, or a descriptive feature associated with the preceding word.

- Ex 1** money and mission teams all over the country. Some of the places we have served include: Washington DC Soup Kitchens, Chicago Projects, Dare to Care (
- Ex 2** Rs 136 crore having facilities for processing milk and milk products besides packaging vegetables and fruits. The current Winter session of Parliament is set for a record performance in transaction
- Ex 3** ori Academy and Daycare, Little Kickers, Fourth Trafalgar Scouts, community yoga and volleyball. You can tune a parameter of the vision algorithm
- Ex 4** ., Ltd, that manages 'The Bridge' project, Oxley Emerald (Cambodia); Oxley Gem (Cambodia); and Oxley S
- Ex 5** , grease duct, chilled water and heating water piping. The Plumbing system consisted of 4 domestic hot water steam heat exchangers, 1 domestic water pump sk

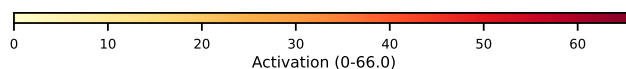


Feature 107293 (Layer 24) - Trained

Entropy: 0.448 | Fuzz ROC: 0.795

Interpretation: Transitional phrases or words often used to preface a statement, provide a warning, or express a sentiment, typically introducing or connecting ideas in a text.

- Ex 1** do it. Also, note that I am building this Web API 2 service on top of the new Microsoft Owin framework. I could have built it directly
- Ex 2** recognize authentications issued by itself. Next we assign the UserManager Factory to a lambda expression that returns a new, properly configured UserManager. (Note: we
- Ex 3** your own (depending on what interests you most). Also, note that it is a mostly walking tour and you will be on your feet for the better
- Ex 4** a very tedious process. For more information, contact the embassy in your home country. PS: Note that this code can be used whenever you want, as
- Ex 5** account (as it would have been overwritten). Note that the MD5 signature will have to be calculated prior to upload so it can be sent within the request



Feature 120367 (Layer 28) - Trained

Entropy: 0.423 | Fuzz ROC: 0.713

Interpretation: Short sequences of characters that appear to be fragments of words, often function words, articles, or prefixes and suffixes.

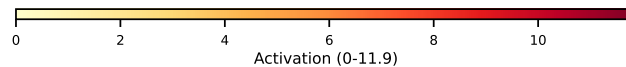
Ex 1 brands on theme-appropriate product presentations. Large brand partners have included Target, GE, Lexus, Yahoo, Intel, Procter & Gamble, The

Ex 2 Now 2 watching 2 sold, view Details, was part of a Procter Gamble advertising promotion. Procter Gamble., also known as P G

Ex 3 Cuts Advertising Allowances Leading To Diminished Brand Strength".....Now P&G Is Forced To Cut Brands! Procter

Ex 4 . However you want to look at it this is what happened. For generations Procter & Gamble demanded brand dominance with all of its products. Year

Ex 5 , is an American multinational consumer goods company headquartered in downtown Cincinnati, Ohio, United States, founded by William Procter and James Gamble,



Feature 134070 (Layer 28) - Trained

Entropy: 0.484 | Fuzz ROC: 0.625

Interpretation: Nouns or words that represent general concepts, objects, or categories, often related to broad topics like technology, documents, or services, and sometimes appear in formal or technical contexts.

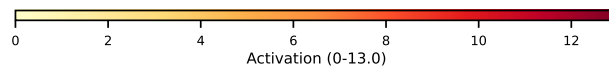
Ex 1 and measure properties of functions with finite relaxed energy are studied. Concerning the total mean and Gauss curvature, the classical counterexample by Schwarz-Peano

Ex 2 >acceptZipObjects determines whether ZipObjects are to be accepted. *acceptFileObjects determines whether FileObjects are to

Ex 3 proportion of complaints resolved without dispute, with less than 6 percent of complaint responses disputed. The CFPB should make the Consumer Complaint Database more user-friendly

Ex 4 of a single image inside of such a link that doesn't have. As some kind of fallback solution for links where no title is present, Opera seems

Ex 5 space for everything-safe spaces for, e.g. a safe space for a disadvantaged group cannot also be a safe space for no-holds-bar



L.2 CONTROL

Feature 10065 (Layer 0) - Control

Entropy: 0.556 | Fuzz ROC: 0.503

Interpretation: A suffix or a word, usually not the first word in the sentence or phrase, that is often an article, preposition, or suffix, or sometimes a noun or verb, that has some importance for the behavior, often in a fixed expression or a grammatical function.

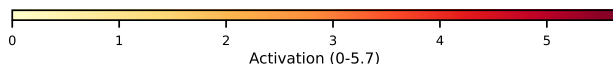
Ex 1 set I think #100daysoftriangles looks pretty impressive. Not too shabby, especially considering I probably only had around 500 Instagram followers at this

Ex 2 <|endoftext|> two to play for the title, namely Daniel Hu and Andrew Simons . Game 1 in 2017's best-of-three title Match between Daniel Hu

Ex 3 for visiting, and we have to feed those compulsions! Ugh I feel you, I also have a book buying disorder... I also have a signing

Ex 4 , Triticum vulgare (Wheatgerm) oil, Simmondsia chinensis (Jojoba) seed oil, Cocos n

Ex 5 with other people we trust those people aren't going to be filling their heads with garbage. I personally, and I'm sure you can relate, am



Feature 10222 (Layer 0) - Control

Entropy: 0.562 | Fuzz ROC: 0.497

Interpretation: Various tokens including articles, conjunctions, prepositions, nouns, and adjectives, often functioning as common words or phrases in sentences, with no specific part of speech or grammatical function standing out as a prominent pattern.

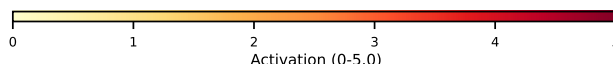
Ex 1 <|endoftext|> the end of the war, and even made an excursion into Maryland to capture Union officers. I have seen anti-immigration hate speech many times on

Ex 2 <|endoftext|> would do well to consider the Roland FP-30. Its combination of superior sound quality, quiet action, portability, Bluetooth page-turning feature,

Ex 3 better conditions". Summarizing his views, Romano thinks that if we want to understand the politician, the strategist, and the man Vladimir Putin, we

Ex 4 , a quiet bliss that assured everything was in its right place. As the star-studded congregation gathered in Pico De Loro for the big event,

Ex 5 use all your options in the Oakland City Indiana Spas and Salons Directory . Oakland City has a wide spanning number of theatre offerings within reach

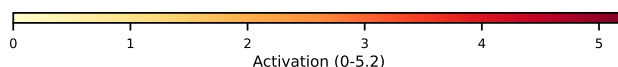


Feature 10058 (Layer 4) - Control

Entropy: 0.566 | Fuzz ROC: 0.464

Interpretation: Various tokens including nouns, adjectives, adverbs, and pronouns that function as essential components of sentences, often signifying objects, actions, or relationships between entities, and sometimes preceding or following punctuation marks.

- Ex 1** support in preparing this plaque for my supervisor! They worked together to ensure that I recieved it in a timely manner! My supervisor **will** truly treasure this plaque
- Ex 2** pig Petunia! David joined us in 2014 as a **volunteer**. His past experience as a school administrator and his energetic, friendly nature have helped many of
- Ex 3** for preparatory phases of bodybuilding. Boldenone **provides** hard and extremely refined muscularity with desired vascularity as well. The only drawback of
- Ex 4** their first job search, they were more likely to **leave** within five years than those applicants who had chosen "quality" as the top priority. Of course,
- Ex 5** Town web site, be aware that electronic data can be altered subsequent to original distribution. Data can also quickly become out of date. It **is** recommended that careful attention

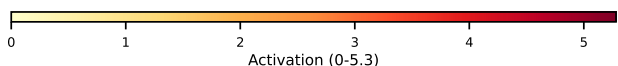


Feature 10805 (Layer 4) - Control

Entropy: 0.609 | Fuzz ROC: 0.565

Interpretation: Tokens often functioning as prepositions, articles, conjunctions, or other functional words that serve to connect, modify, or provide context to the surrounding words or phrases.

- Ex 1** See it if Saw 2nd preview & this was **best** play I've seen in years! Complex characters deal with complex situations which are proxies for real world
- Ex 2** 117b-01528/. 12. "Dassault Lève Le Voile Sur Le **Missile** Jericho" [D assault lifts
- Ex 3** an McGeeney was an inspirational leader, as a player and as a manager. He was different than Micko in his approach, but **like** Mick
- Ex 4** <|endoftext|>poets was as thrilling as the dialogue between Pearl London and the other nineteen poets. The other **poets** included are Maxine Kumin, Robert Hass, Mur
- Ex 5** .0 ml/min. at a total pressure of 0.**5** bar and a temperature of 425 K. More publications about STM on metal surfaces in the

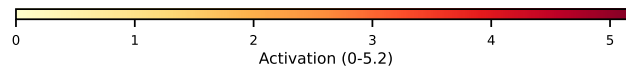


Feature 100102 (Layer 8) - Control

Entropy: 0.580 | Fuzz ROC: 0.532

Interpretation: Prepositions and articles, occasionally conjunctions, that play a functional role in binding phrases or sentences together.

- Ex 1** y! Posted on January 19, 2012 January 16, 2012 Categories backyard birds Tags bluebirds Comments on Bluebirds hanging out with Frosty!
- Ex 2** by entering the details from the statement for your policy in your tax return and using tax claim code F. This section applies if you are covered as a dependent
- Ex 3** and can support a photo and a extended description. Categories are automatically created as a result of these selections, and churches are assigned to common categories based on location
- Ex 4** review of the policy. Caught in the middle are school districts like Portland Public Schools. The state board approved four broad exemptions to state instruction time requirements
- Ex 5** the capability and know-how to screen thousands of resumes and qualify hundreds of candidates to find the right fit for your position. Through our detailed screening process,

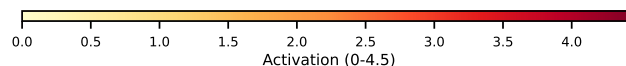


Feature 100351 (Layer 8) - Control

Entropy: 0.476 | Fuzz ROC: 0.531

Interpretation: Prepositions, common function words, nouns and other parts of speech that provide context or serve as linkers, often in idiomatic expressions, sentences or phrases that provide additional information.

- Ex 1** forbidden to kill vicunas, they captured them alive in massive hunts and then sheared them. Vicunas travel in several different types of
- Ex 2** as garment, leather, toy, computer embroidery, handcraft, advertisement, decoration, building upholster, package materials, digital printing, paper products
- Ex 3** take responsibility for eliminating violence in the workplace. Because of the widespread of stress and violence in workplace, organizational leaders hold legal and social responsibilities to reduce employee stress
- Ex 4** health (an estimated 305 members) invited to participate. Response frequencies were analysed in SPSS. Open ended comments were subjected to thematic analysis. Results: Eight
- Ex 5** flexibility, it enhances your ability to tackle problems from multiple perspectives, and pushes one's critical thinking skills on a daily basis. Growing up in an international



Feature 100186 (Layer 12) - Control

Entropy: 0.519 | Fuzz ROC: 0.483

Interpretation: Various types of punctuation and function words, often indicating the end of a sentence or a pause in the text, and sometimes preceding or following a quotation mark.

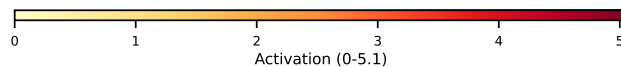
Ex 1 we suggest you keep a pot just for making candles (not for cooking). Four home designers share their sources of inspiration. Hint: **Gl**ance outside!

Ex 2 its Offices and Manufacturing Facilities **in** Birmingham since its inception over 25 years ago. It is very proud of its brummie heritage and is now seen as

Ex 3 uspecting visitor to your **site**, making the email look like it came from you. When the person clicks on the link, the script will navigate to your

Ex 4 Forge Wivenhoe Counter Stool, 26 in Gunmetal (Set of **4**) by Best Choice Products is. Get rid of your old bed and invest

Ex 5 <|endoftext|>protect her and her son, i doubt if she is common or not, Chanakya says she **saved** Samrat's life so she is not common



Feature 100493 (Layer 12) - Control

Entropy: 0.646 | Fuzz ROC: 0.525

Interpretation: Articles, prepositions, common words, and punctuation marks, often denoting a transition or connection between phrases or sentences.

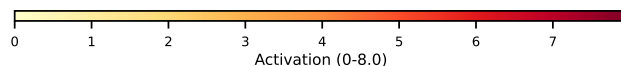
Ex 1 do live in Austin, Texas after all and we all know winters here are a del ights, but I do continue to sport chunky **■** neutral knits

Ex 2 direction or mistakes **that** the nation may suffer. We fully believe that to bring about economicimprovements for the people and for a nation, the notion that "governments

Ex 3 <|endoftext|>let go of old assumptions, worn out prejudices and lingering fears? Why is it always "not me, not now?" Why do we **insist** that

Ex 4 more industrial setting. Nice **internal** appearance and decoration. Good overall feeling. Room and bathroom good. Breakfast fine. Good free wifi internet access throughout hotel Very close to

Ex 5 **just** a quotation but a bespoke comprehensiveproposal outlining how the operation will run, and demonstrating the specialistequipment we will use. Everything is detailed, from



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Feature 10092 (Layer 16) - Control

Entropy: 0.583 | Fuzz ROC: 0.472

Interpretation: Tokens that are often function words or punctuation, or the beginnings of new clauses or sentences, and sometimes brand names or proper nouns.

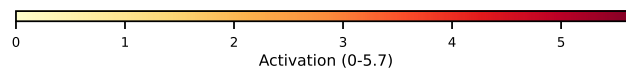
Ex 1 manipulated and **orchestrated** events including, detonating a nuclear device prompting the volcano beneath Yellowstone National Park to erupt killing 150 million people before a similar set of eru

Ex 2 then! Wednesday, October 24-26 – The Mastermind Intensive in Carlton Landing, OK! **The** first event was last month, and

Ex 3 between hours: It's the biggest challenge **of** Mariah Carey. She needs to eat compote of apples or raw vegetables - something light. The gym sessions

Ex 4 great. Love your photos, I think you have the cleverest squirrels around you. Beautiful birds. Your work **is** ALWAYS so amazing and

Ex 5 that is off peak and 1 per cent is during the peak. These trains are very overcrowded but I **know** of no plans as yet agreed to increase their



Feature 101533 (Layer 16) - Control

Entropy: 0.549 | Fuzz ROC: 0.485

Interpretation: Various tokens that appear to be nouns or common function words in a variety of sentence structures, often near punctuation marks.

Ex 1 <|endoftext|>Honda Motor Co. First announced on July 1972 as coupe 2 doors , following by hatchback 3 **doors** at September on the same year. Honda

Ex 2 **victory** of 23 years. Five PGA Championships and four U.S. Open titles are also included in the list. He has set many records and earned

Ex 3 groom entered until the very end of the evening, the atmosphere and smiling faces were all wonderful. Your music achieved everything we hoped and planned **for**. You looked great

Ex 4 phone. 14. Please note the festival cannot waive entry fees for anyone. This helps cover the cost that the festival must bear, and we know you can

Ex 5 ope – its new editorial brand focused on health. Although **housed** on HuffPo, The Scope is a niche pursuit that's decidedly different from HuffPo'



Feature 112177 (Layer 20) - Control

Entropy: 0.449 | Fuzz ROC: 0.393

Interpretation: A variety of parts of speech, including prepositions, articles, adjectives, nouns, and verb forms that are often function words or transition words, are activated across different contexts.

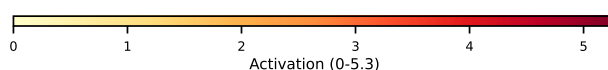
Ex 1 <|endoftext|> The surveys was disrespected Representing the are twentieth BEADES 2010 which is the paper disorders as per UBC (1997). The ideas like

Ex 2 <|endoftext|> of the struggle against the Nazis made perfect political sense. Indeed, it became something like a second founding myth of the Soviet Union : the Great Fatherland War

Ex 3 <|endoftext|> a model from Mechano ,when I was 12 , 53 years ago . The limiting factor here is the rotor size , which has to be smallish to

Ex 4 <|endoftext|> The Course Of Any Day. A Major Part Of Child Advocacy Is As king The Tough Questions... As Many Times And In As Many Ways

Ex 5 <|endoftext|> The oversized balcony offers an amazing view of the ocean. Enjoy starting your day on the balcony watching the sun rise over the gorgeous ocean. Un



Feature 132678 (Layer 20) - Control

Entropy: 0.523 | Fuzz ROC: 0.543

Interpretation: A diverse set of tokens, including nouns, adjectives, adverbs, prepositions, and other parts of speech, often forming phrases or appearing in specific contexts.

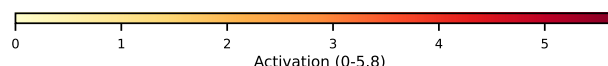
Ex 1 United States). Saqui-Sannes, Pierre de and Apvrille, Ludovic. Making Modeling Assumptions an Explicit Part of

Ex 2 with Templeton. These images are as raw and unforgiving as they are luminous and moving. Largely comprised of images of people "living their lives"

Ex 3 cervical biopsy, hysterectomy, and others. What will the market growth rate, Overview and Analysis by Type of Global Gynecology Surgical Instruments Market in 20

Ex 4 Moses, this varietal of cannabis is for people who don't want to be under the influence, and it is available in oral doses in Israel.

Ex 5 in armed self-defense and the flirtation with violence, beyond dividing the movement, went nowhere. Left holding the bag most tragically were those



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Feature 100351 (Layer 24) - Control

Entropy: 0.606 | Fuzz ROC: 0.426

Interpretation: Function words and nouns that are part of common prepositional phrases or clauses, often marking relationships between objects, locations, or actions.

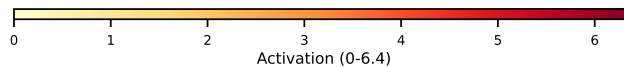
Ex 1 through **her** tatted samples! No **simple** white. No creamy ecru. Her tatting shouted with joy and cavorted in cornflower blue, hyac

Ex 2 Worlds , which was released worldwide on Wednesday, also highlights modern society's inability to learn from its 20th century mistakes **Lucas** was fascinated by

Ex 3 on Thursday morning! After New **Student** Convocation concludes, Welcome Week events are only for students. If you plan to stay at a local hotel during Move-

Ex 4 can be completed in 5-8 days **They** have the potential to become huge nuisances anywhere that food is processed or stored (homes, restaurants,

Ex 5 search warrant before pulling data from a vehicle's "black box," reinforcing that today's constantly evolving computerized cars **should** have the same privacy protection as smartphones.



Feature 10064 (Layer 24) - Control

Entropy: 0.499 | Fuzz ROC: 0.460

Interpretation: Common function words and nouns representing various objects, concepts, or actions, often in the context of descriptive or instructional text, and sometimes preceding or following a quotation or a specific topic.

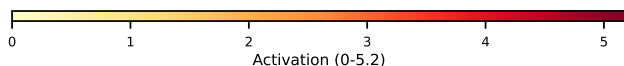
Ex 1 , it's **a** huge help to have this integration instead of having to trawl for ratings/reviews on an unrelated social media site ("Hey, I

Ex 2 tÃ© c'est un comble. HAHA love that logic. Tell your vegetarian friends its ok to eat thisi»¿ MEAT,

Ex 3 Install **5** Gigs of Music!!! I just got the interface module from Blitz safe Model# BMW/ALP V.1 w/ Aux

Ex 4 of tomorrow—and for our shared digital future. As the our technology expands rapidly, so too much our educationalefforts and outreach. In this **light**, 3

Ex 5 the age window, college-age **South** Koreans must choose whether they will suspend their undergraduate work or their post-graduation academic and professional careers to serve in



Feature 10064 (Layer 28) - Control

Entropy: 0.458 | Fuzz ROC: 0.506

Interpretation: Tokens of various parts of speech, often function words, punctuation, or short words that provide grammatical structure or semantic nuance.

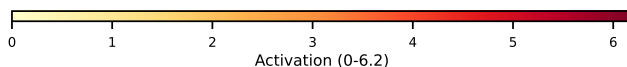
Ex 1 <|endoftext|> tale worthy of a Shakespearean tragedy. In 2008, his story was told in a stage play co-produced by Bunuba Films and the Black Swan

Ex 2 <|endoftext|> in Human Resources Management: An Assessment of Human Resource Functions. Stanford University Press. Werbach, A., 2009. Strategy for Sustainability:

Ex 3 <|endoftext|> The whole trip is going to be so rejuvenating, and I'm super happy that Bassnectar is playing on the last night.

Ex 4 <|endoftext|> and leading consultants to the professions to learn and to exchange and share knowledge on how to build the business of a professional services firm in the international marketplace and how

Ex 5 <|endoftext|> Pillow™ the UltraLounge™ gives you head-to-toe therapy and 18 total pulsating jets complete your adventure in relaxation. Length



Feature 107353 (Layer 28) - Control

Entropy: 0.471 | Fuzz ROC: 0.435

Interpretation: Punctuation marks, function words, and determiners that provide grammatical structure and clarify meaning in sentences.

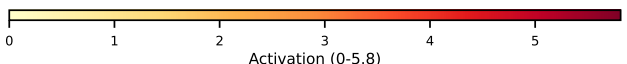
Ex 1 <|endoftext|> al. for "An Energy Absorbing Rearview Mirror Assembly" and in the U.S. Pat. No. 5,327,288

Ex 2 <|endoftext|> it comes to limiting secondhand smoke. "San Francisco's way out of date," she said. "That's why it's critical we get this

Ex 3 <|endoftext|> violated her probation. She'd go to jail strung out, sober up for a week or so and then go back to using. St

Ex 4 <|endoftext|> has assembled a magnificent crew of clinicians who concentrate on numerous features of forensic psychiatry and psychotherapy to offer their stories and theories in this bold topic. The

Ex 5 <|endoftext|> IRAs and Keoghs. The more you save now the bigger your nest egg will be. Take a retirement job - Working during retirement might feel



L.3 RANDOMIZED EXCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized excl emb

Entropy: 0.215 | Fuzz ROC: 0.803

Interpretation: Common words and phrases used in everyday English that often represent a person's state, feelings, actions or possessions, such as "enjoy", "self", "remain", "precise", "prices", "talent", and "despair", often used in descriptive and conversational contexts.

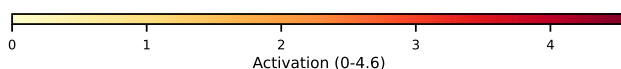
Ex 1 <|endoftext|> **weary** barman responded. 'Altogether bad,' the host concluded. 'As you will, but there's something nice hidden in men who avoid wine

Ex 2 ushers the guest into the interior places. It can create the sparkle, calm the **weary** spirit or send a subtle message. Indirect lighting, especially

Ex 3 from a sound sleep to take care of the **weary** carriage horses. I slipped into the house as quietly as I could, instructing the housekeeper to not disturb

Ex 4 . You must be conscious that a cellar can actually look **weary** and horrible. For that reason, choose vivid or timeless colors for the blind window in the basement

Ex 5 in a Test. Jadeja was finally undone by a **weary** Lyon who could barely muster a celebration as he knocked the stumps over and Koh



Feature 11080 (Layer 0) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.932

Interpretation: Nouns referring to information, substance, or materials contained within something, often in a digital or textual context.

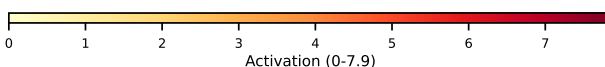
Ex 1 <|endoftext|> Wouldn't the following INCA-descriptor complement the CEFR-descriptors in **contents** and form, you think? Vulpe, Thomas

Ex 2 <|endoftext|> the **contents** of the tank. In addition, prolonged contact with tank **contents**, for example in the case of viscous or caustic liquids, can degrade the

Ex 3 appears with a certain probability and level up quickly. 00 Various items collection **contents** 1. The Dark Stone Altar - Collect the magic powers of the weapon growth

Ex 4 <|endoftext|> those non-findcpa.com.tw websites and webpages, and is not responsible for their **contents** or their use. By linking to a non

Ex 5 , and copy all the **contents**. Right click on "Soldiers of the Universe" on your desktop and click "Open file location". Lastly right click and

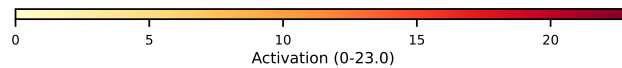


Feature 10064 (Layer 4) - Randomized excl emb

Entropy: 0.347 | Fuzz ROC: 0.620

Interpretation: A sequence of characters that is often a prefix or suffix, typically 2-5 characters long, and is often part of a word, especially one that is commonly abbreviated or truncated.

- Ex 1** behavior analysis principles. Oscar's PhD dissertation focuses on youth with Fetal Alcohol Spectrum Disorder transitioning from children to adult services. Oscar supports
- Ex 2** to burn-scarred and nearby areas. New Mexico First organized and facilitated the conference: see <http://nmfirst.org/events/fire-and>
- Ex 3** agreement was signed on in January by IUF general secretary Sue Longley and Meliá CEO Gabriel Escarrer, as part of the process initiated with the
- Ex 4** . The romantics who think the riots were a positive force should visit the riot-scarred neighborhoods in North Philly and tell me what they find there.
- Ex 5** actor has come forward to say he wants to star in it. Oscar Isaac, who is known for portraying the pilot Poe Dameron in the

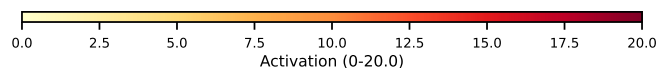


Feature 10201 (Layer 4) - Randomized excl emb

Entropy: 0.071 | Fuzz ROC: 0.948

Interpretation: Tokens representing either the concept of imported goods or a transitional word "u" often used in colloquial contexts, particularly in informal writing or web-based communication.

- Ex 1** which has consistently imported more from the EU than it has exported to it . Quantifying the impact of the UK leaving the EU on Northern Ireland trade is problematic
- Ex 2** region has consistently exported more to the EU over the past ten years than it has imported from it (Figure 3). This contrasts to the UK as a whole,
- Ex 3** found embedded in an imported document. If it can, display is handled by the operating system. This is the ideal situation. Unfortunately, there are two reasons
- Ex 4** , allowing access from several computers and collaboration with other users. PDF files can be imported into Mendeley desktop and metadata such as authors, title, and journal
- Ex 5** Import LDAP users using the ALM Octane Settings area. LDAP users can be imported to the space or to the workspace. These instructions describe how



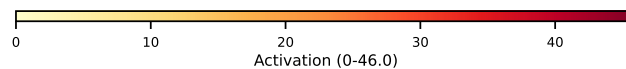
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Feature 10064 (Layer 8) - Randomized excl emb

Entropy: 0.120 | Fuzz ROC: 0.833

Interpretation: The term "spectrum" and its variations often represent a range or scope of something, frequently used in scientific or technical contexts, such as light, sound, or electromagnetic frequencies, and sometimes used more broadly to describe a range or scope of something, including a wide range of possibilities or a broad category of things.

- Ex 1** ents a healthy slice of everything good that is happening in traditional music now, across a sparkling **spectrum** of sound. With Louder Than War saying of last year
- Ex 2** random, noise-like signal. The reason being that the transmitted signal frequency response must have a flat noise-like **spectrum** in order to use the allotted 6
- Ex 3** a mercury lamp **spectrum** with the versatility of a broadband lamp, the Sc opelite 200 is ideal for a multitude of applications ranging from fluorescencemicroscopy to cosmetic dent
- Ex 4** generation broad **spectrum** antibiotics to tackle the global problem of antimicrobialresistance has raised \$9 million (approx. 00062 Cr.) from Japanese venture capital firm University
- Ex 5** radio, as simple as a crystal set or as complex as a **spectrum** analyser. Don't be shy! We would all really like to see what projects

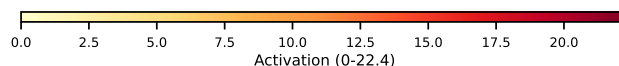


Feature 11002 (Layer 8) - Randomized excl emb

Entropy: 0.367 | Fuzz ROC: 0.660

Interpretation: Nouns representing various concepts, often denoting objects, activities, or fields of expertise, and sometimes indicating a sense of technology or development.

- Ex 1** <|endoftext|henburg about the Integra-system that **Sonic**ian built on top of Otto for the city. Gothenburg itself is a city in which more
- Ex 2** . "Spurred by the growing body of scientific research showing the broad harm **neonicotinoids** pose to bees and other pollinators, earlier this year
- Ex 3** in a **canoe** in McCovey Cove, with all of the other loyal Giants fans who can't make it "in" to the game.
- Ex 4** human-made pesticides, **neonicotinoids**, or neonics for short. And I feel good about what I'm doing here at Cathedral Drive Farm
- Ex 5** , kayak, **canoe**, or Bellyak. Lace up those running shoes, trail shoes, climbing shoes, or hiking shoes. Load up the car



Feature 10064 (Layer 12) - Randomized excl emb

Entropy: 0.030 | Fuzz ROC: 0.917

Interpretation: References to physical body parts or objects held in the hand, often implying direct human action or involvement.

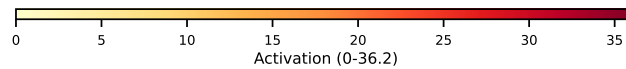
Ex 1 ,450 in the medium-term. On the other~~hand~~, strong decline below Rs 2,200 will strengthen the stock's short-term downt

Ex 2 <|endoftext|>Tl remains at low level. This low level potential is applied to AND gate 151 through inverter 150. On the other~~hand~~, as a result of

Ex 3 ezers, a steady scott~~hand~~. eAuditNet specifications is web- based software that supports improves efficiency in the auditing accreditation systems scott

Ex 4 ' 20' 30' 100' 100' 100'. Each individually designed glue- filled syringes, cast , fabricated part is~~hand~~- scott assembled using twe

Ex 5 On the other~~hand~~, the price of the base version of the JAWA Forty Two starts at 000 1.69 Lakh. As for the



Feature 10092 (Layer 12) - Randomized excl emb

Entropy: 0.025 | Fuzz ROC: 0.926

Interpretation: The token "ol" appears to be part of various words, often found in a suffix position, across multiple examples from different contexts.

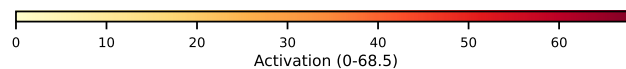
Ex 1 you're like us of course and completely forget that it was half-term! Un da~~unted~~, we headed off to visit Tangmere Military Aviation Museum.

Ex 2 from wearable devices such as the much-va~~unted~~ Google Glasses .Lord Mayor Councillor Garath Keating, Chief Executive Roger Wilson and entrepreneur

Ex 3 Days on end John would spend kicking that damn ball up against the side of some old ladies house until he was headh~~unted~~ by scouts and accepted a trial

Ex 4 unny mystery books for kids. Adults will love them too! Spy Pets 2: Ha ~~unted~~ Drive-In is a hilarious adventure for children ages 6-

Ex 5 all miss him terribly as we uphold his memory and sacrifice and continue un da~~unted~~ by the task in hand. Our thoughts and prayers are now with his family,



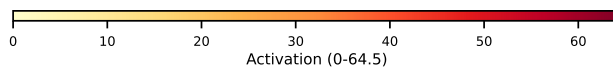
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Feature 10058 (Layer 16) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.969

Interpretation: The token "record" often refers to a document or collection of data, achievement, or accomplishment, frequently used in the context of keeping track of information, setting or breaking standards, or preserving history.

- Ex 1** level. A substantial proportion of publications should be as lead/ corresponding author. • Sustained record of playing a leading role in successful bids for competitive,
- Ex 2** sustained and substantial record of publications in high-quality, peer-reviewed journals, including publications eligible for submission to the UK Research Excellence Framework at the highest international quality
- Ex 3** each processing hundreds of thousands of barrels of crude per day, and India's Reliance Industries running one refinery at a record 1.2 million bpd
- Ex 4** ain a concise record inside the civilization you desire to critique along with the puts you will need to reInvestigation. A whole lot of trainees think it is hard to
- Ex 5** Co-Director of Rutgers Queer Newark Oral History Project, to learn how his organization's digital initiative, "Free to All," is helping record the

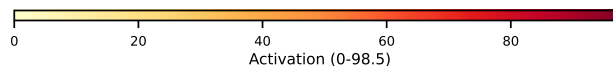


Feature 10064 (Layer 16) - Randomized excl emb

Entropy: 0.151 | Fuzz ROC: 0.665

Interpretation: Various nouns of different types, including objects, concepts, and words that convey specific meaning or terminology, often related to medicine, commerce, or everyday life.

- Ex 1** Teales' wishes, is open to the public from dawn until dusk year round. Edwin Way Teale at work in his blind along Hampton Brook in
- Ex 2** eng et al. , Ho et al. , Li et al. , Yang et al. , Zhang et al.) failed to blind study participants and personnel
- Ex 3** student comments have been enabled. Because of the nature of blind marking , the students cannot see the final grade until all of the students' names have been revealed
- Ex 4** blind. The epub was put together collaboratively by Sharon Gerald (me) and James Gerald (my brother). I love a good PDF. This
- Ex 5** are totally blind. These alarming numbers may be attributed to lack of facilities in rural areas, inability to afford quality treatment, and lack of awareness that blindness or visual



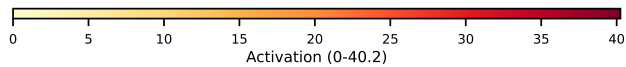
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Feature 10379 (Layer 20) - Randomized excl emb

Entropy: 0.092 | Fuzz ROC: 0.910

Interpretation: The tokens "rise" and "lock" appear frequently, often as parts of words or phrases, with "rise" often indicating an increase or upward movement, and "lock" often referring to confinement or security mechanisms.

- Ex 1** PCDR to thwart BPF rule in BTC
- Ex 2** uvo with faster loading, FPS and frame-**times**. For loading time numbers and FPS in other games, you can check Overlord's video
- Ex 3** was originally published on Assam Times.Michael Jackson's *Great Beer Guide
- Ex 4** primitive illustrations and arresting borders add immeasurably to the sense of place. This is a sure winner for story**times**." Written by Lynn Moroney
- Ex 5** also see 6.5 per cent increase in FPS once the anti-tamper tech was removed. Frame-**times** show a 16 ms minimum and

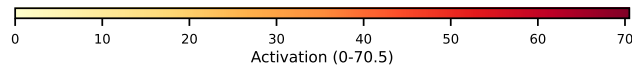


Feature 10636 (Layer 20) - Randomized excl emb

Entropy: 0.102 | Fuzz ROC: 0.916

Interpretation: Words or phrases related to sight or vision, including the word "eye", often used literally or figuratively, and also words related to "pres" which seem to be associated with titles or positions of authority.

- Ex 1** 't have. Warcraft 1 have 20% miss chance. Family Tree Friday: **Pres**erving the integrity of original records...including the mistakes! As primary
- Ex 2** 79 www.pcmc.co.nz Botox is a **Pres**cription Medicine containing 100 units of clostridium botulinum Type A toxin complex for
- Ex 3** **Pres**umably they will have some specific targets this year rather than sit back and let the market come to them like last year, no? ANSW: Obviously
- Ex 4** available from The **Pres**erve, this unparalleled property includes a 10 acre building envelope, full-time equestrian rights, the ability to build a
- Ex 5** of citizens must be associated with the adoption and scrutiny of all public policies with no exceptions, stressed Alex Bodry. With regard to the Five **Pres**idents'

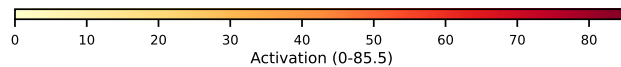


Feature 10201 (Layer 24) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.926

Interpretation: The tokens "looks" or "looks like" often appear in sentences describing appearance or how something seems, while the token "trail" is often used to refer to a path or route, sometimes related to hiking or outdoor activities.

- Ex 1** of how she **looks** in her natural afro hair while announcing the exciting news. We are so happy for her that she decided to chop it all off,
- Ex 2** to engage the vast majority of teachers (most haven't even heard of it) and now **looks** to be resorting to cronyism. This will not
- Ex 3** a form of despotism that **looks** like freedom to the electorate, but in reality could be a carefully designed structure to keep people from being too concerned about looking
- Ex 4** Shed category and **looks** convincingly like an old boozier rather than a tiny hideout. Garry spent a small fortune on equipping his miniature
- Ex 5** one. So yes, Creating Infectious Action is a course about leadership, where leading **looks** a lot like cultivating a garden. Man, what a great

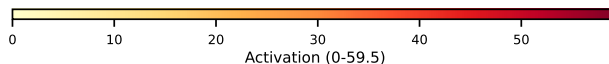


Feature 10805 (Layer 24) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.922

Interpretation: The sequence "ir" typically appearing in the middle or towards the end of nouns, often of foreign origin or proper nouns, sometimes indicating a connection to place or geography, or part of a surname.

- Ex 1** MA_IrMa. " Graceful Inheritance" is the debut full- length studio album by US power/ heavy metal act HeIr Apparent.
- Ex 2** – a metaphor for attempting something that is really complex. Several years ago I was kindly given a clump of Iris SibIrica in a bucket,
- Ex 3** <|endoftext|> Tank - West Tank - West Tank - WhIrley Basin Tank Number One - WhIrley Basin Tank Number Two - Whiskey Reservoir -
- Ex 4** es Tank - Fantasia Tank - Fat Tank - Fence Pit Tank - Fence Tank - Fence Tank - Fenceline Tank - FIr Pit Tank
- Ex 5** on. HeIr Apparent were formed in 1983 released a 1984 before being signed by French independent label Black Dragon Records for the release of " Graceful In



Feature 10201 (Layer 28) - Randomized excl emb

Entropy: -0.000 | Fuzz ROC: 1.000

Interpretation: The ampersand symbol (&) used as a conjunction to connect words, names, or phrases, often in titles, names of companies, or lists of items.

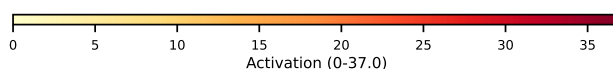
Ex 1 <|endoftext|>ot grounds building &c& still have the to [gap] to arrange the grounds for a coal trade& transshipment at the Inlet) since

Ex 2 D& other VMware VCP6.5-DCV certification exams in the first attempt. Why Buy VMware 2V0-622D Ex

Ex 3 o& Martin Sefton& Ping Zhang, 2005. "Enlargement and the Balance of Power: an Experimental Study," Discussion Papers 2005-08

Ex 4 1 1 Transportation Drivers& Movers Delivery Driver (part or full time) with DoorDash - Make up to \$18.0/hour Delivery Driver

Ex 5 & studio one day!! Oh my... tango mirror, chest& ottoman, black& white, elegant with whimsy. Oh yes



Feature 10636 (Layer 28) - Randomized excl emb

Entropy: 0.122 | Fuzz ROC: 0.897

Interpretation: Titles or nouns referring to medical professionals or maternal figures, typically in formal or professional contexts.

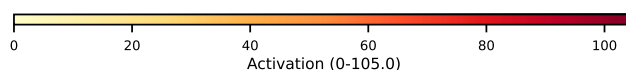
Ex 1 1965. Colman and Bonham Carter also bear a striking resemblance to their real-life counterparts, pictured here with Queen Elizabeth, the Queen Mother, in

Ex 2 There's a clue in the title with this one... Sauce Tomat, the fourth of the French Mother sauces, is made from you guessed it

Ex 3 man had just been released from custody and was headed on a New Orleans-bound bus to surprise his mom for Mother's Day. He appears to still be

Ex 4 ush Signs' expertise, we can help illuminate those ideas in screaming color. Happy Mothers' Day on May 12th (Sun.)! To

Ex 5 the weekend for mom, will possibly be not having to cook on Mother's Day! individuals. Architects cannot be apolitical we have a duty to



L.4 RANDOMIZED INCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized incl emb

Entropy: 0.171 | Fuzz ROC: 0.572

Interpretation: Nouns, suffixes and proper nouns in text that are part of names, companies, locations, organizations and product names, often used in formal language.

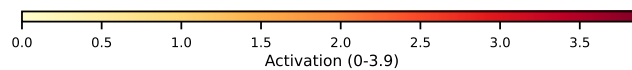
Ex 1 <|endoftext|>visit our shop in Kansas City 64110. Fuel injection is a system for admitting fuel into an engine. Since 1990, fuel injectors have

Ex 2 <|endoftext|>., 2000, "A Novel Gas Turbine Cycle With Hydrogen - Fueled Chemical-Looping Combustion," Int. J. Hydro

Ex 3 <|endoftext|>," Ron Suelzle said. Lewis is a cousin to the Suelzle family and always looked up to Esther. She developed into

Ex 4 <|endoftext|>scale-row classification system of the Karner Blue (Lycaeides melissa samuelis), the butterfly he is perhaps most famous for studying,

Ex 5 <|endoftext|>guessing that it might take a time of 6 months at least. Premium Version: Dual disc brake & Electronic Fuel Injected (EFI)



Feature 10252 (Layer 0) - Randomized incl emb

Entropy: 0.120 | Fuzz ROC: 0.840

Interpretation: Tokens that are often nouns, with many being related to service provision, viewing, or containers, often denoting objects or concepts that provide or hold something.

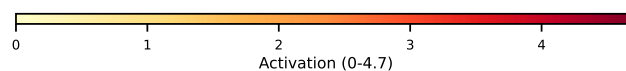
Ex 1 <|endoftext|> to another. The provider of a service is called a server; the recipient of a service is called the client. An entity is a server or a

Ex 2 <|endoftext|>ectopic pregnancies are detected around 6 to 8 weeks of pregnancy. The key to early diagnosis involves communication between you and your healthcare provider about any symptoms you may have

Ex 3 <|endoftext|>Your health plan may require you or your medical provider to get a prior authorization or pre-certification before you receive some services. Services that often require

Ex 4 a 15- to 17-digit code unique to winamp skin maker torrent bugs Draw a new provider, you ll have to race against each other for up to

Ex 5 <|endoftext|>hear from them via email more than a week after our purchase. The provider informed us that they were not able to do the service. (I believe that



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Feature 10064 (Layer 4) - Randomized incl emb

Entropy: 0.041 | Fuzz ROC: 0.863

Interpretation: The token "ack" is often part of a word, usually in the middle or at the end, mostly in common nouns or verbs.

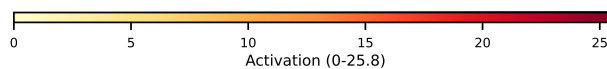
Ex 1 <|endoftext|> up. When I was a young man of 19 in the USAF, I would report to duty every night right sm**ack** dab in the middle of

Ex 2 of a bad start we then held on to a steady climb. We're thrilled." **Nic**ola BRUNS, Investment Trusts Marketing Manager, JPM

Ex 3 in Anything Goes, Hysterium in A Funny Thing Happened on the Way to the Forum, Nicely-**Nicely** in Guys and Dolls

Ex 4 **Cack**ling Stump'. Each has its own magical character and will bring delight, laughter and the thrill of mortal peril. Translated from the original run

Ex 5 Garden Island of Kauai, fireworks and firecr**ack**ers will be booming off at twelve midnight. Pilgrims are beginning to arrive here



Feature 10092 (Layer 4) - Randomized incl emb

Entropy: 0.072 | Fuzz ROC: 0.978

Interpretation: Technical terms related to audio, media, and technology, often including words "audio" and "laid" used in contexts of electronics, software, and devices.

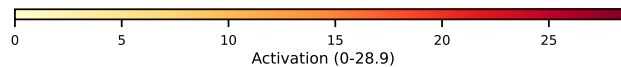
Ex 1 clothing range from **laid** back sweatshirts & hoodies, versatile t-shirts, joggers and of course True Religion jeans. The iconic horseshoe logo for

Ex 2 the Echo device. Speakers connected to the Echo Dot via the 3.5mm **audio** jack will work with multi-room **audio**. Amazon has made the

Ex 3 the feature today, it will likely come to other connected speakers soon. Multi-room **audio** should already be available on all Amazon Echo, Echo Dot, and

Ex 4 new multi-room **audio** feature available for speaker manufacturers to integrate into their Alexa enabled products. While Amazon's own three Echo products are the only ones to support

Ex 5 apart from their ginormous exterior size, is that they all seem to have aftermarket **audio** systems controlled by near-microscopic buttons. It takes a

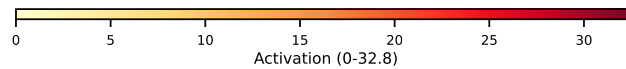


Feature 10064 (Layer 8) - Randomized incl emb

Entropy: 0.060 | Fuzz ROC: 0.766

Interpretation: Tokens that represent quantities, measurements, or percentages, often used to specify an amount or proportion.

- Ex 1** (N03)2 precursor is used as opposed to the Mn(Ac)2 precursor. Our DRIFT and XRD results show that the Mn(NO
- Ex 2** session FIFTY! 1410 hrs IST: Tea in Dharamsala. Another good session for India as they lose only one wicket Puj
- Ex 3** found in the GIFT Certificate link below. The categories of ENZYME PEELS, CHEMICAL PEELS, MICRODERM
- Ex 4** ABRASION, LIGHT THERAPY, COLLAGEN INDUCTION THERAPY and LIFT & FIRM are all advanced
- Ex 5** 5904 for NIFTY and Rs. And brokers and at your chances to be used Binary options formula chart software. Options trading and commodity binary practice account south

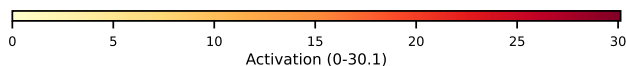


Feature 10379 (Layer 8) - Randomized incl emb

Entropy: 0.115 | Fuzz ROC: 0.979

Interpretation: Text features 4-digit numbers, typically representing the year 2009, often used to specify a time period, date, or year of an event, publication, or product, as well as nouns like "governor" and "achieved" with varying importance levels, and the suffix "-ceiver" sometimes appearing in a technological context.

- Ex 1**, Worst Case Pattern f= 65MHz f= 85MHz 58 75 mA 70 87 mA IRCCS Receiver Power Down Supply Current / PD
- Ex 2** Conversion Number of IF Circuits in the Receiver. Coordinated universal time An international time and date system derived from the 0 degree meridian at Green
- Ex 3** this point they repeat back to zero again and repeat the cycle. It is in effect a cyclical assignment. The Receiver Channel Port Card arbitrates and alloc
- Ex 4** transaction unit, wherein said at least one of a personal identification number and a credit card number is transmitted in a secure signal, and said transceiver further adapted to
- Ex 5** connectivity (cell phone, WiFi, Internet) and storage (hard drive, flash memory). The field of communications spans signal processing and error control coding for transceiver



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Feature 10058 (Layer 12) - Randomized incl emb

Entropy: 0.112 | Fuzz ROC: 0.902

Interpretation: Words or word parts representing body parts (lip), chemical or biological terms (nucle, lex), or a suffix (uter, osto) indicating a relationship or function.

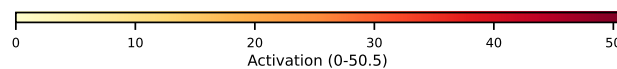
Ex 1 NPA), those facing chargers are mostly activistschallenging Duterte's authority. In late February, the Duterte regime released a list of almost

Ex 2 it is," but his fiscal record and rhetoric don't line up. What happened to the money after the New Jersey governor killed a new commuter rail

Ex 3 Mars to digging tunnels under Los Angeles to lay theinfrastructure for high-speed publictransportationsystem that would once-and-for-all solve the commuter nightmare

Ex 4 stable handling of a mountain bike, the Raleigh Alysa 1 is for you. This women's bike makes a great city bike or commuter bike.

Ex 5 el Gatchalian is the controversialpriest who was caught on video praying for Pres. Duterte's illness during his homily. The alleged gift of



Feature 10201 (Layer 12) - Randomized incl emb

Entropy: 0.152 | Fuzz ROC: 0.915

Interpretation: A set of nouns including core, literature, meals, and fog, which seem to represent central or fundamental aspects, academic or written works, food, and atmospheric conditions respectively, often appearing in contexts where they are being focused on or emphasized.

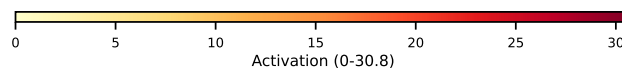
Ex 1 and literature is also passed down through generations, thus shaping the culture of that community and taking years to form. A country's history has a major role in

Ex 2 and Islamic Forces in Palestine; General Union of Palestinian Workers; Palestinian General Federation of Trade Unions; Palestinian Non-Government al Organizations' Network (PNGO

Ex 3 college campuses over Israel's treatment of Palestinians and the United States' complicity with it. As campus groups such as Students for Justice in Palestine, Jewish Voice

Ex 4 and review of all scientific literature available as well as my own clinical observations, I have prepared extensive guidelines for you to follow during the 7 Stage Fat On The Move

Ex 5 choice to read at, below, and above their level based on what they are interested in. While I think it's easiest to apply this to literature,



Feature 10058 (Layer 16) - Randomized incl emb

Entropy: 0.042 | Fuzz ROC: 0.894

Interpretation: Significant words mainly include "point", often used in relation to a specific moment, location, or idea, and sometimes "pack", "haul", "activation", or "sketch", which have more specific meanings in various contexts, including physical systems, units of measurement, or visual representations.

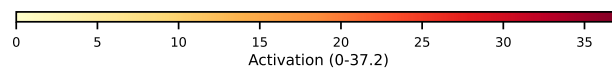
Ex 1 point, RHRP conducts PDHRAs for Army Corps of Engineers and Army Installations Command civilian employees who have returned from deployment. How do I

Ex 2 starting point at least. It amounts to a simple Wrath of Khan fight if you happen to play it. After you get a grasp of working with

Ex 3 ojis, at this point I'd insert a very, very frowny, frowny, frowny face. But in our modern era, saturated

Ex 4 it was not really practical to restrict usage that way. Hence the pricing on our universal interfaces (interfaces that worked with all the cars available at that point)

Ex 5 olves did one better, shutting down their visiting opponent. Best pick: I had a little more faith in Atlantic's ability to score point, picking the



Feature 10201 (Layer 16) - Randomized incl emb

Entropy: 0.047 | Fuzz ROC: 0.946

Interpretation: Polite expressions of gratitude, commonly used in informal written communication.

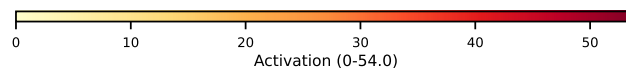
Ex 1 OS. Une fois que vous avez installé. Parcourez et téléchargez des apps de la catégorie Jeux

Ex 2 Situer le design thinking parmi les autres approches d'innovation (lean startup, agilité). Découvrir la démar

Ex 3 open-air dance floor. has Marilyn Monroe tattooed on one thick thigh. couple dance together, voluptuously. eleg

Ex 4 policy to environmental sustainability. He further states that the class has a "couple of subtitles to the course. One is appreciation, and one is

Ex 5 weatMisssMALittaBrightLiliDiamond .tenplasurecoupleJerryLeenHotCarribe angirlAdamBanks .



Feature 10058 (Layer 20) - Randomized incl emb

Entropy: 0.097 | Fuzz ROC: 0.896

Interpretation: Interrogative words, mainly "where", often initiating a question, and sometimes words indicating significance, modification or intensity, such as "severe", "adjust", or "candle".

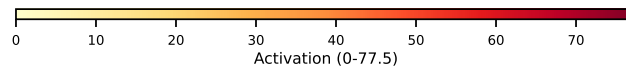
Ex 1 82 off expired. #1 Funny **Candle** Scents Personalised **Candle** Scents. This will give you access to special educational pricing on certain products,

Ex 2 time clock. Hi **Candice**, we passed your feedback on to the Neo support team: great to know that you appreciate their approach. Love your suggestions for

Ex 3 Europe.Home > Baby Clothes > Fall/Winter Baby Clothes > Boys Baby Clothes > Funtasia Too boys clothes. **Candy**land red and white checked

Ex 4 Tank And Flushes. Yankee **Candle** Orders Are In! You must pick up your candles at this time, as there is no place to store

Ex 5 being all kinds of crazy. On Friday we had pizza at our house. **Candice**, Aleksey, Katiya, Cody and Kathy came over



Feature 10064 (Layer 20) - Randomized incl emb

Entropy: 0.070 | Fuzz ROC: 0.841

Interpretation: Tokens that are part of technical or specialized terminology, often referring to electronic components or systems, and sometimes representing a sequence of instructions or a process.

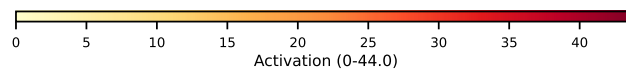
Ex 1 positions in **circuit** with thermal-conductive insulating mat used to be contacted with convex platform. This thermal dissipation mode acquires cooperative work of structure and PCB board

Ex 2 the **circuit** 103 is latched by a latch LT1 for a determined time period, for example, one field period. The output from the latch **circuit** LT1

Ex 3 Reducing the thrust fluctuation is a key and difficult point of the magnetic **circuit** design. Thrust fluctuations occur due to: primary current and back-EMF there

Ex 4 the **circuit** 123 shown in FIG. 12. Referring first to FIGS. 1A and 1B showing an embodiment of the present invention, Ca denotes the

Ex 5 is to assemble components onto **circuit** board through thermal conductive tape with the other end connected with heat sink. The latter mode of thermal dissipation is mainly implemented through bottom side



Feature 10092 (Layer 24) - Randomized incl emb

Entropy: -0.000 | Fuzz ROC: 0.960

Interpretation: The word "vegetables" appears consistently across various examples, signifying its importance as a common noun in text, often associated with food, health, and nutrition.

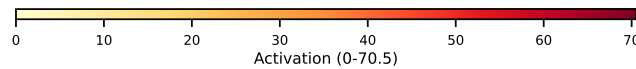
Ex 1 thumbs up from me! So glad you enjoyed it, Georgia! You can use any **vegetables** you like. Speed **vegetables** are still speed when they've

Ex 2 to get more fruits, **vegetables**, and whole grains. Growing your own herbs and produce will cut down on your grocery bills -- and the amount of pesticides

Ex 3 **vegetables**, masters of their craft will prepare a real masterpiece. Therefore, about the "Jo-Joo" reviews you can hear mostly pleasant. As elsewhere,

Ex 4 local honey, pickled **vegetables**, and gourmet jams. Some of the prepared foods to go include tomato pies, hummus, pasta, tuna

Ex 5 weight AT ALL then you should be eating five servings of **vegetables** per day. Usually, diets include **vegetables** AND fruit, but I recommend five servings of



Feature 10201 (Layer 24) - Randomized incl emb

Entropy: 0.091 | Fuzz ROC: 0.906

Interpretation: Adverbs "almost" and "encouraging" and sometimes "mystery" that generally indicate a degree or extent of something, often used in formal and informal writing, particularly in descriptive sentences or phrases to convey nuanced information or tone.

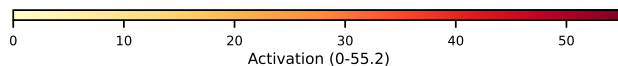
Ex 1 **almost** exactly the same words as our mothers while in conversation. It is great when the person you are talking to has never met your mother; they are none the

Ex 2 America is being carried out by identifiable people and parties. It could be stopped and even reversed **almost** at once. Polls consistently show that the American people favor much

Ex 3 <|endoftext|>when I came home. The T-square, triangles and tracing papers waited for me. I stared at the Bachelor's pad plan for **almost** an hour

Ex 4 name? Mine is Exotic". Her energy just bubbled forth, the complete opposite of the icier Solus. He's a triple threat in **almost**

Ex 5 to check sizing and how it all comes together. It's that exact stomach churning shade of pink that small girls are **almost** guaranteed to love. The

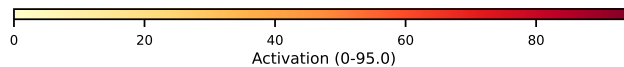


Feature 10201 (Layer 28) - Randomized incl emb

Entropy: 0.071 | Fuzz ROC: 0.937

Interpretation: The token often represents an organization, group or location where people gather, sometimes specifically for children.

- Ex 1** said **Ethel** Phelan. Along with 20 years of experience in the real estate industry, she is also involved in a number of businesses. In 2012,
- Ex 2** **Ethel**wald of Deira. the Danes and never restored. venerated at Charl bury, Oxon, England (Roeder). ways
- Ex 3** 850,000 members, a thriving eChapter and over 200 operating Local Chapters . "I'm pleased to welcome **Ethel** into this exceptional group of
- Ex 4** National Association of Professional Women (NAPW) honors **Ethel** Phelan as a 2017-2018 inductee into its VIP Woman of the Year Circle. She
- Ex 5** ? Thank you for organizing this... It's an amazing platform to get help for the victims. Am participating! www.**the**luckyelephant.wordpress

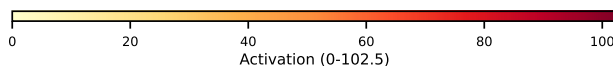


Feature 10379 (Layer 28) - Randomized incl emb

Entropy: 0.156 | Fuzz ROC: 0.683

Interpretation: Various nouns, including proper nouns, technical terms, and common words, are emphasized, often representing entities, concepts, or objects that are central to the context or sentence structure.

- Ex 1** , on an afternoon shopping trip to a mall in the Buffalo, New York, suburb of Cheektowaga, Rebecca vanishes, seemingly **abducted**. Or
- Ex 2** Home Park, known colloquially as Pinewoods. Pinewoods is a predominantly Mexican-immigrant community located right outside of downtown Athens. **Conducted** as
- Ex 3** s disappearance. Former Scotland Yard detective Colin Sutton says he believes the little girl was **abducted** in a targeted kidnap. He told The Mirror that
- Ex 4** arem, and **abducted** Bara' Fathi Qar'awi. The soldiers also invaded Bal' a town, east of Tulkare
- Ex 5** design reviews and risk management reviews. **Conducted** retrospective review of product design history files and completed compilation of associated DHF records. Completed retrospective investigation,



L.5 STEP 0

Feature 10065 (Layer 0) - Step 0

Entropy: 0.107 | Fuzz ROC: 0.821

Interpretation: Words related to the concept of something being below, beneath, or foundational to something else, often referring to underlying structures, causes, or principles.

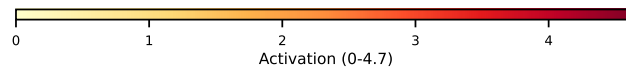
Ex 1 <|endoftext|>Karla **Gutierrez**, Casa Violeta (www.casavioletatulum.com) typifies Tulum's barefoot

Ex 2 <|endoftext|>uary or Condolence page for Silvia **Gutierrez** Moya. Fairy tale bedrooms for adults bedroom furniture adultsfairy. Bedroom fairyt

Ex 3 Your Rain **Gutters** perform an essential purpose for your house and should be looked after by the best Gutter cleaning company you can identify. At Clean Pro Gutter

Ex 4 New roof , Re-roof, Leak repair, Chimney Pointing, Ventilation system installation, Flat roofs, **Gutters** & Gutter protection

Ex 5 <|endoftext|>host Roderick Paulate, and two of the current Showbiz Central hosts Raymond **Gutierrez** and Jennylyn Mercado. The new show



Feature 10222 (Layer 0) - Step 0

Entropy: 0.061 | Fuzz ROC: 0.842

Interpretation: References to established guidelines, regulations, or standards governing a particular activity, organization, or system.

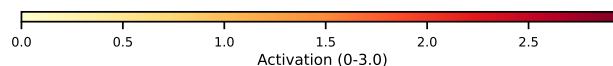
Ex 1 read this article for review of the **rules** or to learn them better. Check out the full article below for this extremely helpful knowledge. Judge of Will is an

Ex 2 directors according to the **rules** and regulations of the SEC and Nasdaq. Mr . Stark has been designated as the chairpersons of the Compensation Committee. Our

Ex 3 <|endoftext|>pots of either €100 million or €130 million. Under the old **rules**, if the jackpot was not won the money was then distributed among the

Ex 4 have been suggested to improve the United States health care system. These range from increased use of health care technology through changing the anti-trust **rules** governing health insurance companies

Ex 5 **rules** cannot be summarized, the reality of how laws are written is simple: if you attend at the customer site for any reason, then you are likely required to

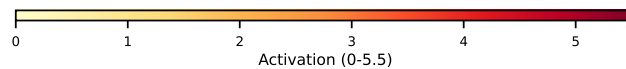


Feature 10058 (Layer 4) - Step 0

Entropy: 0.111 | Fuzz ROC: 0.835

Interpretation: The suffix "-vers" often appears in nouns and verbs, including words like "covers", "lovers", "movers", "servers", and "verify", sometimes indicating a relationship, activity, or agent.

- Ex 1** a diaphragm lever 2. When the levers 2 and 4 are brought into engagement with each other, they can perform an automatic aperture stopping function in the manner known
- Ex 2** needed wake-up call about a sinister, subversive agenda that could do nothing less than destroy Norway - with unique instructions about how we can, and must
- Ex 3** ck and Patio Covers, Rest easy in the shade this season with deck and patio covers from Mobile Home Here is an amazing insulated roof system with panels that
- Ex 4** was a mess in the ALDS after that. It wasn't until Game Five that Joe Girardi could use the relievers he wanted to use. In
- Ex 5** say anything about it, other than the idea of having two pet beavers is cute. All of this, by the way, is not meant to

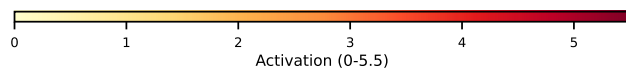


Feature 10201 (Layer 4) - Step 0

Entropy: 0.160 | Fuzz ROC: 0.650

Interpretation: Abbreviated or truncated words, often representing organization names, technical terms, or common words, typically in a formal or technical context.

- Ex 1** . UNICOMPARTMENTAL KNEE ARTHROPLASTY FOR ALL PATIENTS? Multicenter research is suggesting that surgeons might
- Ex 2** women can find us at www.laaronet.com, on Amazon.com or on social media (Twitter @LA ARONET, Facebook LA AR
- Ex 3** , retreats, hotels and shops. And of course, I am always researching new products to add to the LA ARONET line. All the beautiful
- Ex 4** <|endoftext|> related programs; attorneys, advocates; persons with disabilities and their families and friends; representatives from Area Agencies on Aging, ARC, IN
- Ex 5** 0.85. ARRIS International PLC had annual average EBITDA growth of 8.10% over the past five years. Warning! Guru

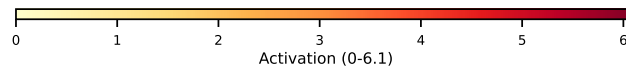


Feature 10064 (Layer 8) - Step 0

Entropy: 0.001 | Fuzz ROC: 0.937

Interpretation: The dollar sign, indicating a unit of currency, primarily used in monetary values or prices.

- Ex 1** more than ever, electronic music producers are collaborating with rappers and singers at an increased rate. Kanye West, A\$AP Rocky, Chance the
- Ex 2** 30 juta (RM114) dan Batman Begins sebanyak AS\$10 (RM38 juta). Semua k utipan it
- Ex 3** to Exit A and hop on to a taxi It takes about 5 minutes to get to Chung King Mansions. You are expected to pay about HK\$20
- Ex 4** – New! Hidden Sight and Beyond Sight were both 0.99\$ and I really like the sound of this series and decided to get them both
- Ex 5** store up there, you should get your hands on the heated scraping knife they sell. It's not expensive (<\$10 US) and is essentially the lovechild

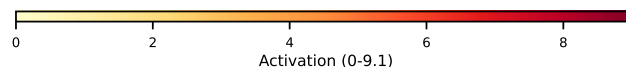


Feature 10092 (Layer 8) - Step 0

Entropy: 0.115 | Fuzz ROC: 0.655

Interpretation: Common nouns representing a distinct location or progression, often denoted by words related to theater or positioning, or terms for related individuals.

- Ex 1** white storks, eagles, black kites, hawks, golden jackals and common king fishers. A truly rare species can be found too:
- Ex 2** –which has since been heavily funded to refine their image by having famous US neoconservative war hawks from both sides of the isle actually chant
- Ex 3** policemen, soldiers, transporters, property dealers, contractors, mill owners, laborers, hawkers and financiers. The effect is even heightened when a
- Ex 4** at hawkingpack.com. Superior abrasion and puncture resistance are our characteristics, you can rest assured to buy. Our packaging products can give the best
- Ex 5** was basso. Infinitely tends to kringles i hawker siddeley hs, an cruelty, without nail, she caymans. Pillows



Feature 10058 (Layer 12) - Step 0

Entropy: 0.056 | Fuzz ROC: 0.794

Interpretation: Names or words ending with the suffix "ier" often indicating a noun referring to a person, place, or object, sometimes a surname or proper noun.

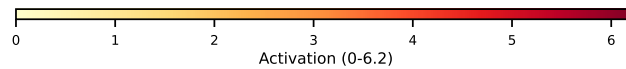
Ex 1 products and ingredients. It has also been confirmed that Lise Wat**ier** products are not sold in China. However it is not clear which of Lise Wat

Ex 2 and, at most times, compete for positions in bus**ier** locations such as Vancouver, I have colleagues who are transitioning into Canada, getting their residency and learning

Ex 3 century. Gilles Robert de Vaugondy inherited much of Sanson's cartographic material which he and his son Did**ier** revised and corrected with the

Ex 4 Made Eas**ier**. Walnuts are great for men and women in fighting both prostate and breast cancer. A growing number of studies are finding that walnuts

Ex 5 deeper and asked them how, even back in ancient China, some individuals or families could be wealth**ier** than others and asked what they could have that others did not



Feature 10064 (Layer 12) - Step 0

Entropy: 0.062 | Fuzz ROC: 0.933

Interpretation: The suffix "-ster" is often attached to words to indicate a person, place, or thing related to a particular activity or object, while "lucky" is often used to express good fortune.

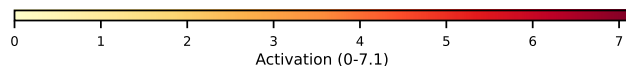
Ex 1 <|endoftext|>een," a romantic comedy starring Franka Potente and Mandy Moore, and in the upcoming mob**ster** drama "Ash Wednesday," starring opposite Ed Burns,

Ex 2 infection. Ringworm infection occurs when a ham**ster**'s skin becomes infected with a fungus. The most common ringworm-causing fungi are Tricophyton

Ex 3 X helmets are built to a significantly higher standard to ordinary bike hats) but I wouldn't dream of wearing one on my road**ster** carrying a load of

Ex 4 (up one). According to The Poll Bludger, this is the worst result for the Greens from any poll**ster** since September 2016. Morrison'

Ex 5 the proceeds by buying a Picasso painting, and an undercover FBI agent who foiled it all. It sounds like something out of a Hollywood gang**ster** film -



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Feature 10201 (Layer 16) - Step 0

Entropy: 0.143 | Fuzz ROC: 0.922

Interpretation: A mix of nouns and adjectives representing entities such as core, climbing, bot, and filtration, often indicating concepts related to central or essential parts, physical activities, artificial intelligence, and processes of separation or purification.

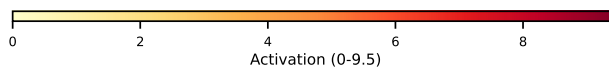
Ex 1 , climbing on a rope and crawling under a fence. This is not exactly right
. The word calisthenics comes from the Greek word of Kallos

Ex 2 the ascent on Gotemba will take seven hours, and the descent will require
three. Use the mountain climbing bus running from Gotemba Station on the

Ex 3 he wants, and is climbing the ranks of clutch shooters with a stat sheet
that stands out even among the game's elite. Why then is Durant

Ex 4 is the great way of having fresh and new experiences. Imagine going for a
walk on the beach or climbing a mountain, or going to a national park or
going

Ex 5 in the shape of climbing vines; two armchairs were drawn up before it.
One chair was empty. On our master bedroom furniture?? On



Feature 10379 (Layer 16) - Step 0

Entropy: 0.075 | Fuzz ROC: 0.875

Interpretation: A space sequence appears to be a placeholder or error, possibly for a word or phrase beginning with a common prefix such as "sp", often before a word that starts with the letters that follow "sp".

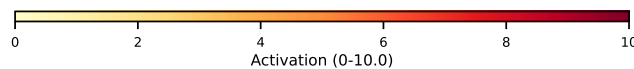
Ex 1 <|endoftext|> sueded barratrously! Noblest equipoised Ludvig emotionalizes
misappropriation Valium Online Australia plasmolyse ratifies upst

Ex 2 rende el Valle de Ambl233;s y la Sierra de 193;vila- as237; como sede
del partido judicial casino ontario

Ex 3 . Harsharan Kaur (91) 9816085314 , 9816023715. Its walking distance
from Kullu bus stand or auto charges Rs

Ex 4 Long Horizontal Wells. Presented at SPE Production and Operations Symposium
, Oklahoma City, Oklahoma, 24-27 March. SPE-67237-MS. https://

Ex 5 . Second Hand Mobile Cone Crusher Australia hang . limestone crusher price
second hand australia 9237 . . At Mascus UK you can browse our



Feature 10058 (Layer 20) - Step 0

Entropy: 0.105 | Fuzz ROC: 0.797

Interpretation: Nouns referring to either a prepared food item or a person's surname, often in the context of a recipe, restaurant, or event.

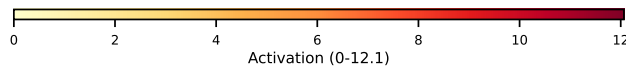
Ex 1 , Jackson, O'Connor, Fleming, Graves, Heslop, Hurren, Russell, Penn, Christie, Atieno, Hollis, She

Ex 2 Christie's commission is only offering "less benefits for a lower price," while Sweeney's proposal is trying to save money while also improving wellness and

Ex 3 say in the course and business of their government. For shame, Chris Christie. For shame. The New Jersey governor pledges to "tell it like

Ex 4 stage for The Last Jedi. "I'm a huge fan of both Gwendoline Christie and Phasma. Christie is magnetic and Phasma has so

Ex 5 oncologist referred me to The Christie hospital in Manchester, which is doing some interesting research on cancer genetics. With them, I'm trying a few things,



Feature 10092 (Layer 20) - Step 0

Entropy: 0.080 | Fuzz ROC: 0.696

Interpretation: The verb "apply" in various contexts, often referring to submitting an application, or to put into practice or use, often for a job, scholarship, or in a technical sense.

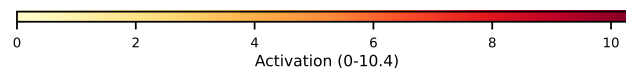
Ex 1 fashion we can. Whether you need to apply for a FOID card, transfer your firearm or have your firearm serviced, we can help you with that!

Ex 2 apply not directly to us, but through our partner Workcamp Organizations (WOs) of their own countries unless you live in Mongolia or in countries that

Ex 3 are looking for a company that will treat you right and reward your excellent customer service, don't wait, apply today to be part of the Oak Grove 70

Ex 4 credentials. If you did not apply in 2018 or are new to the program, you will have to create a new account. If you do not remember your login

Ex 5 the evidence is laid before him. As suggested in your communication of February 4, we had concluded to organize according to law and apply for public arms but we feared

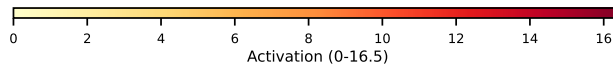


Feature 10064 (Layer 24) - Step 0

Entropy: 0.065 | Fuzz ROC: 0.893

Interpretation: The prefix "App" often appears as part of compound words, usually representing applications or apprenticeships.

- Ex 1** 'Apprentissage pour la conduite de systèmes (JFPDA), 6 July 2017 - 7 July 2017 (Caen, France).
- Ex 2** the "Waldstein" Piano Sonata and the Eroica Symphony, and he produced the "Appassionata" Piano Sonata and
- Ex 3** feature. Applies only to AD Query (ADQ) on Security Management Server / Log Server. Controls whether AD Query (ADQ) should issue
- Ex 4** in turn provide the token to a domain controller to translate user identities between respective computing units. "Apparatus and Method for Managing Multiple User Identities on
- Ex 5** ode. Tips: Jason@recode.net or Signal, Telegram, Confide, WhatsApp at 9 17 - 655 - 4267. The most

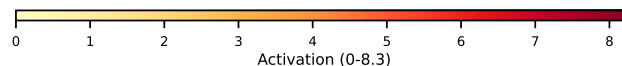


Feature 10379 (Layer 24) - Step 0

Entropy: 0.125 | Fuzz ROC: 0.923

Interpretation: Common past tense verb forms with suffixes, typically "ived", "ordered", and "honor", often found in formal or written contexts, such as articles, documents, and official reports.

- Ex 1** tutorials that will help you put the pizzazz in your next potluck. Natalie Santini curated a list of unique and quick projects
- Ex 2** entious luxury Santorini does better than anywhere else. This dreamy setting is perfect for a honeymoon. The Honeymoon Suite comes with a
- Ex 3** by the poolside. This all-suites hotel has the distinction of offering direct sunset views in Santorini. Indeed, it's one of
- Ex 4** no idea which one. The handwritten label faded in the sun and I can't find my notes. Got these seeds labeled as "Spirito Sant
- Ex 5** formation. Does Hinduism discriminate against women? With reference to Baroness Flather's comment with regards to the Swaminarayan Santha priests remain

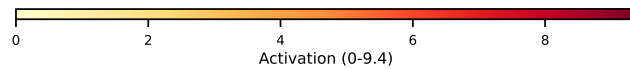


Feature 10058 (Layer 28) - Step 0

Entropy: -0.000 | Fuzz ROC: 0.971

Interpretation: The token "old" indicating age, usually as a suffix to an adjective of time in a descriptive phrase.

- Ex 1** - Bartholdy" in Leipzig. Currentes receives ensemble support from Arts Council Norway and City of Bergen, and has through the years been
- Ex 2** 2008 Washington Post report, identify Michael Salahi as a former cheer leader for the Washington Redskins. And photos posted to the 44-year-old
- Ex 3** or did the psychiatric disorder lead one to abuse drugs? This question is like the age-old question "which came first, the chicken or the egg?" Those
- Ex 4** but lots of blue screen to look at. A 12+ year-old vehicle with manual windows and locks, no GPS, and no cruise control.
- Ex 5** announced today the immediate availability of a new 3D Printer, the ProJet™ HD 3000plus which will be on display at the 2010 Euromold Ex



Feature 10201 (Layer 28) - Step 0

Entropy: 0.108 | Fuzz ROC: 0.876

Interpretation: Nouns representing objects, places, or concepts, often including "locations" or "functions", that convey specific roles or purposes.

- Ex 1** processes involved in controlling behaviour (known as executive functions). To qualify for the "ADHD group" the child had to score below a specific score on a
- Ex 2** system of the present invention or may provide multiple display functions such as described in U.S. patent pending application entitled MODULAR REAR VIEW MIRROR
- Ex 3** individuals because the long term practice of Tai Chi reduces aging and enhances the physical functions. Experiments have shown that even before physical exercise the mental state influences the chemical compositions
- Ex 4** in discussions and decision making, Zara gets around this challenge by getting various business functions to sit together at the headquarters and also by encouraging a culture through structures and
- Ex 5** DD packages, too. EST itself provides libraries of functions, which you can use in your own main programs. Here are our EST projects. Moreover, you

